Lecture 14 – Return-oriented programming

Stephen Checkoway

Oberlin College

Based on slides by Bailey, Brumley, and Miller

ROP Overview

- Idea: We forge shellcode out of existing application logic gadgets

- Requirements:
  vulnerability + gadgets + some unrandomized code

- History:
  - No code randomized: Code injection
  - DEP enabled by default: ROP attacks using libc gadgets published 2007
  - ROP assemblers, compilers, shellcode generators
  - ASLR library load points: ROP attacks use .text segment gadgets
  - Today: all major OSes/compilers support position-independent executables

Image by Dino Dai Zovi

ROP Programming

1. Disassemble code (library or program)
2. Identify *useful* code sequences (usually ending in ret)
3. Assemble the useful sequences into reusable *gadgets**
4. Assemble gadgets into desired shellcode

* Forming gadgets is mostly useful when constructing complicated return-oriented shellcode by hand

A note on terminology

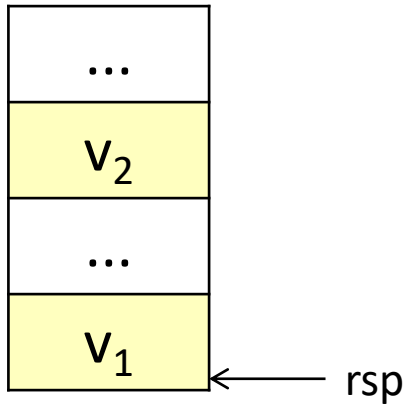
- When ROP was invented in 2007
  - Sequences of code ending in ret were the basic building blocks
  - Multiple sequences and data are assembled into reusable gadgets
- Subsequently
  - A gadget came to refer to any sequence of code ending in a ret
- In 2010
  - ROP without returns (e.g., code sequences ending in call or jmp)

There are many *semantically equivalent* ways to achieve the same net shellcode effect

Equivalence

```
Mem[v2] = v1
```

Desired Logic



Stack

```
mov rax, [rsp]
mov rbx, [rsp+16]
mov [rbx], rax
```

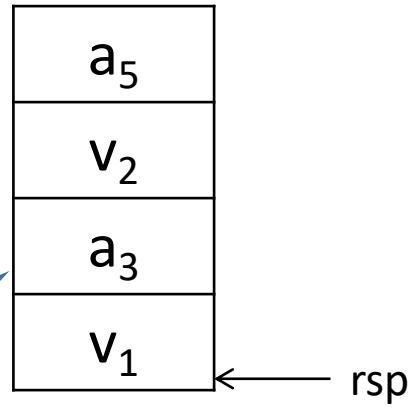
Implementation 1

Constant store gadget

`Mem[v2] = v1`

Desired Logic

Suppose a_5 and a_3 on stack



Stack

rax	v_1
rbx	
rip	a_1

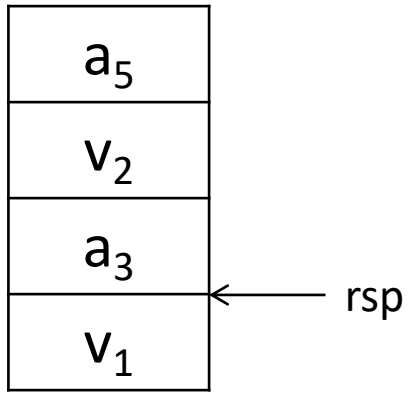
```
a1: pop rax;
a2: ret
a3: pop rbx;
a4: ret
a5: mov [rbx], rax
```

Implementation 2

Constant store gadget

```
Mem[v2] = v1
```

Desired Logic



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Implementation 2

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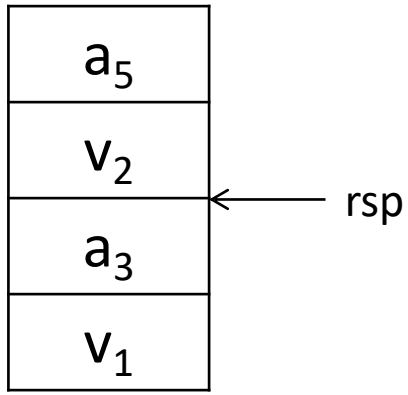

Stack

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rbx	v_2
rip	a_3

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a1: pop rax;
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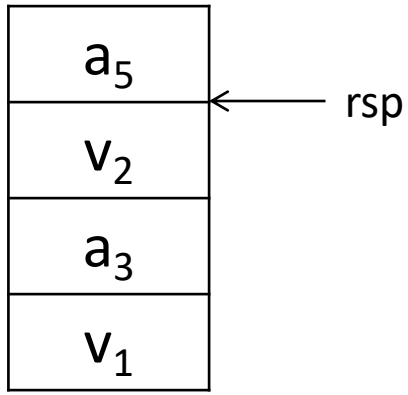
Implementation 2

Constant store gadget

```
Mem[v2] = v1
```

Desired Logic



Stack

rax	v_1
rbx	v_2
rip	a_4

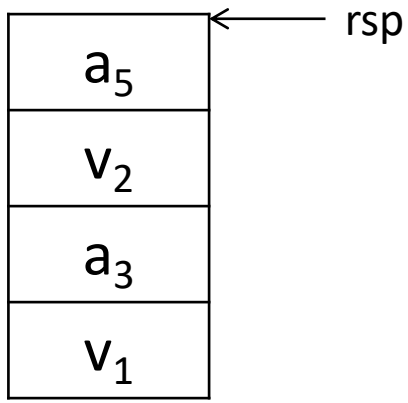
```
a1: pop rax;
a2: ret
a3: pop rbx;
a4: ret
a5: mov [rbx], rax
```

Implementation 2

Constant store gadget

```
Mem[v2] = v1
```

Desired Logic



Stack

rax	v_1
rbx	v_2
rip	a_5

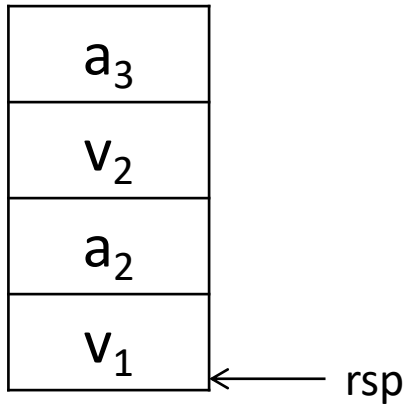
```
a1: pop rax;
a2: ret
a3: pop rbx;
a4: ret
a5: mov [rbx], rax
```

Implementation 2

Equivalence

`Mem[v2] = v1`

Desired Logic



Stack

semantically equivalent

a_1: `pop rax; ret`

a_2: `pop rbx; ret`

a_3: `mov [rbx], rax`

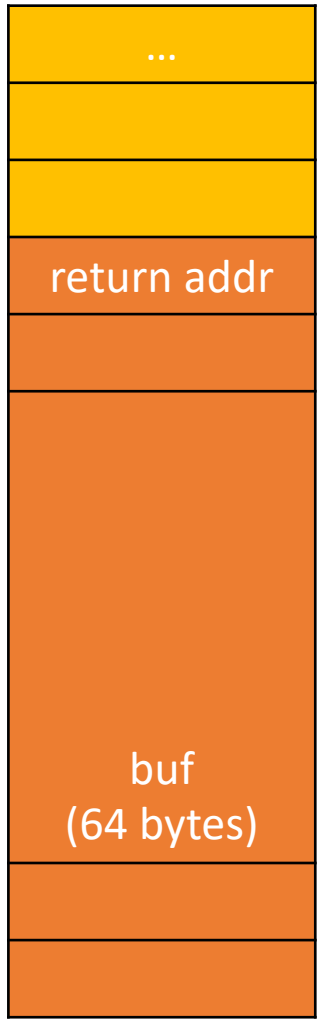
Implementation 2

Return-Oriented Programming

```
Mem[v2] = v1
```

Desired *Shellcode*

- Find needed instruction gadgets at addresses a_1, a_2, and a_3 in *existing* code
- Overwrite stack to execute a_1, a_2, and then a_3



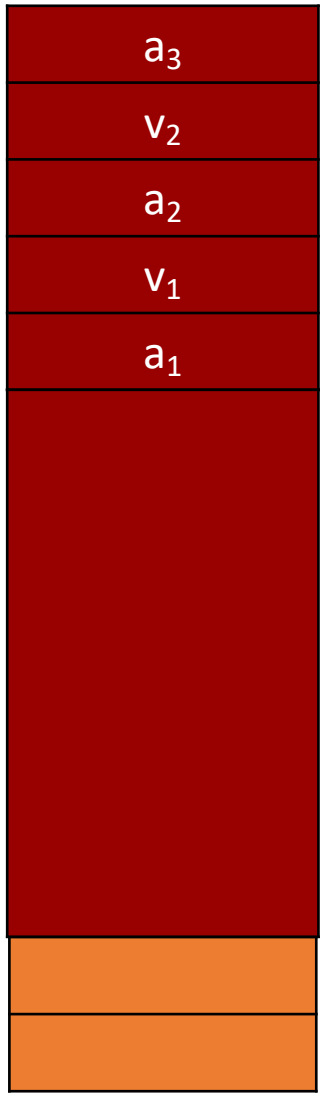
Return-Oriented Programming

`Mem[v2] = v1`

Desired *Shellcode*

```
a1: pop rax; ret
a2: pop rbx; ret
a3: mov [rbx], rax
```

Desired store executed!



Arithmetic/logical operations: c = x op y

Basic strategy

1. Pop the address of one variable into a register
2. Load the value of the variable into a register
3. Pop the address of another variable into a register
4. Load the value of the variable into a register
5. Perform the operation
6. Pop the address of the destination variable into a register
7. Store the result of the operation at that address

Must be mindful of register interactions

Arithmetic

- Addition: c = x + y
  - pop rax
    ret
  - mov rax, [rax]
    ret
  - mov rbx, [rax]
    ret
  - add rbx, rax
    ret
  - mov [rax], rbx
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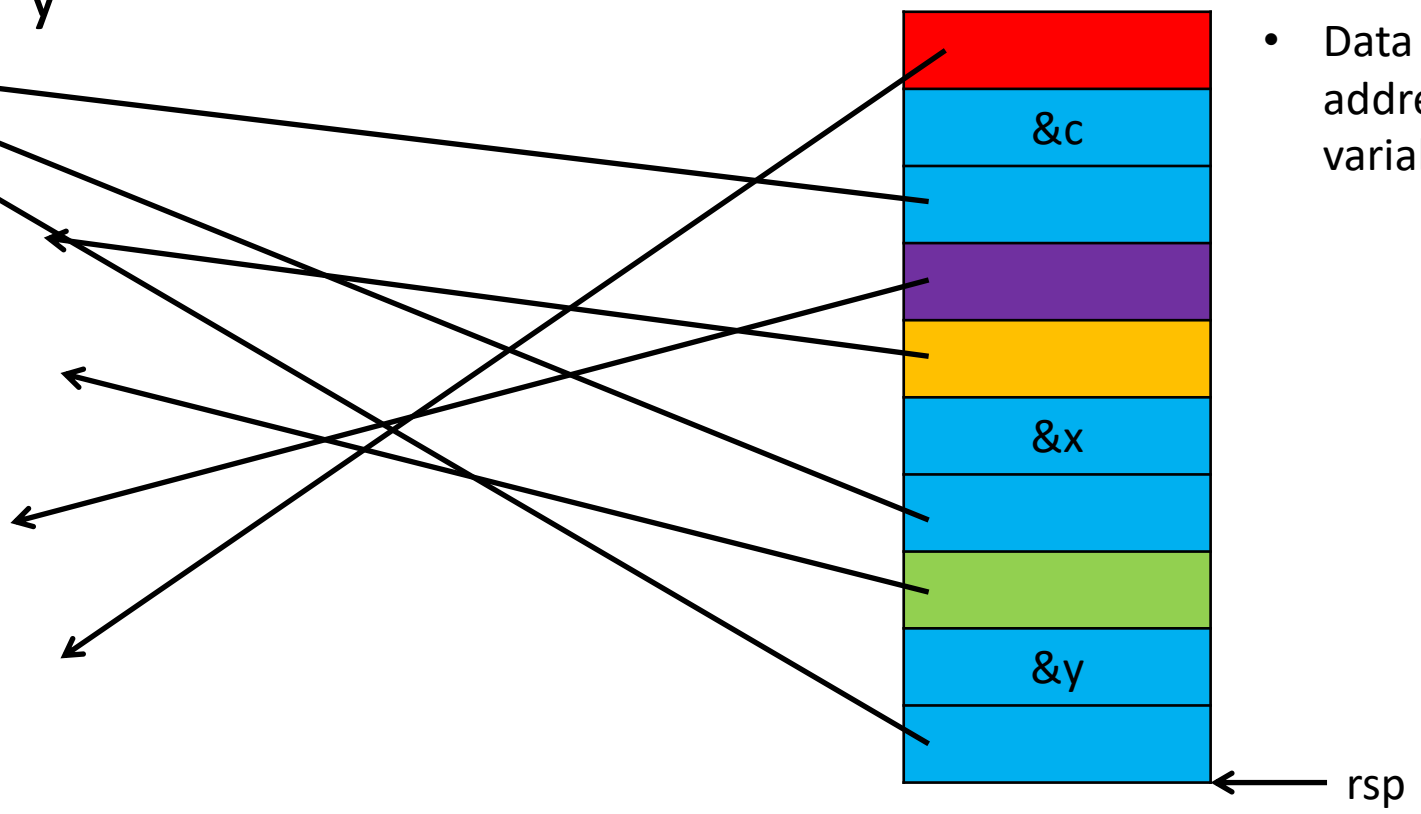

Stack contains
- Addresses of code snippets ending in ret
- Data (here, the addresses of our variables)

Arithmetic

Register	Value
rax	105
rbx	3852

- Addition: c = x + y
  - pop rax
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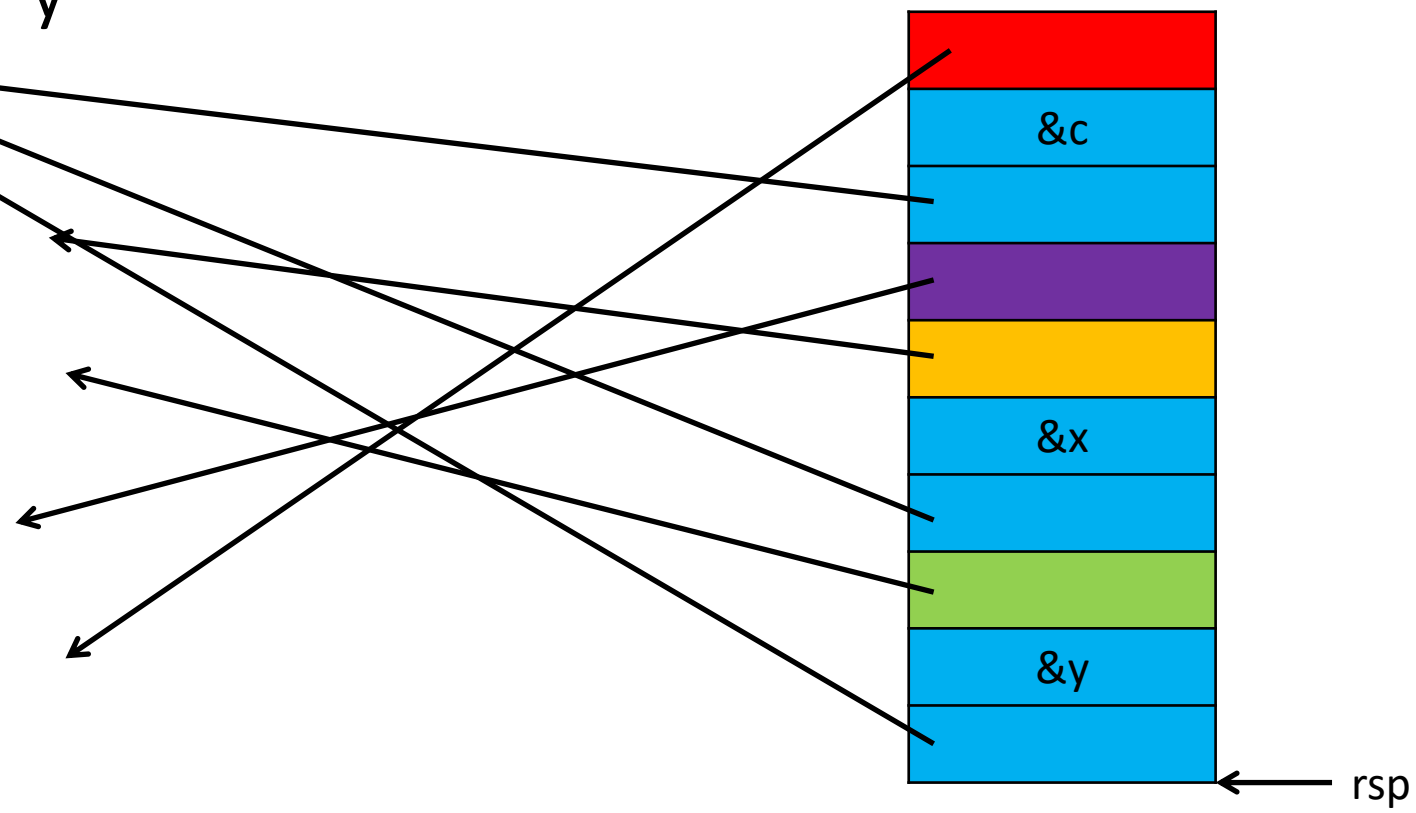

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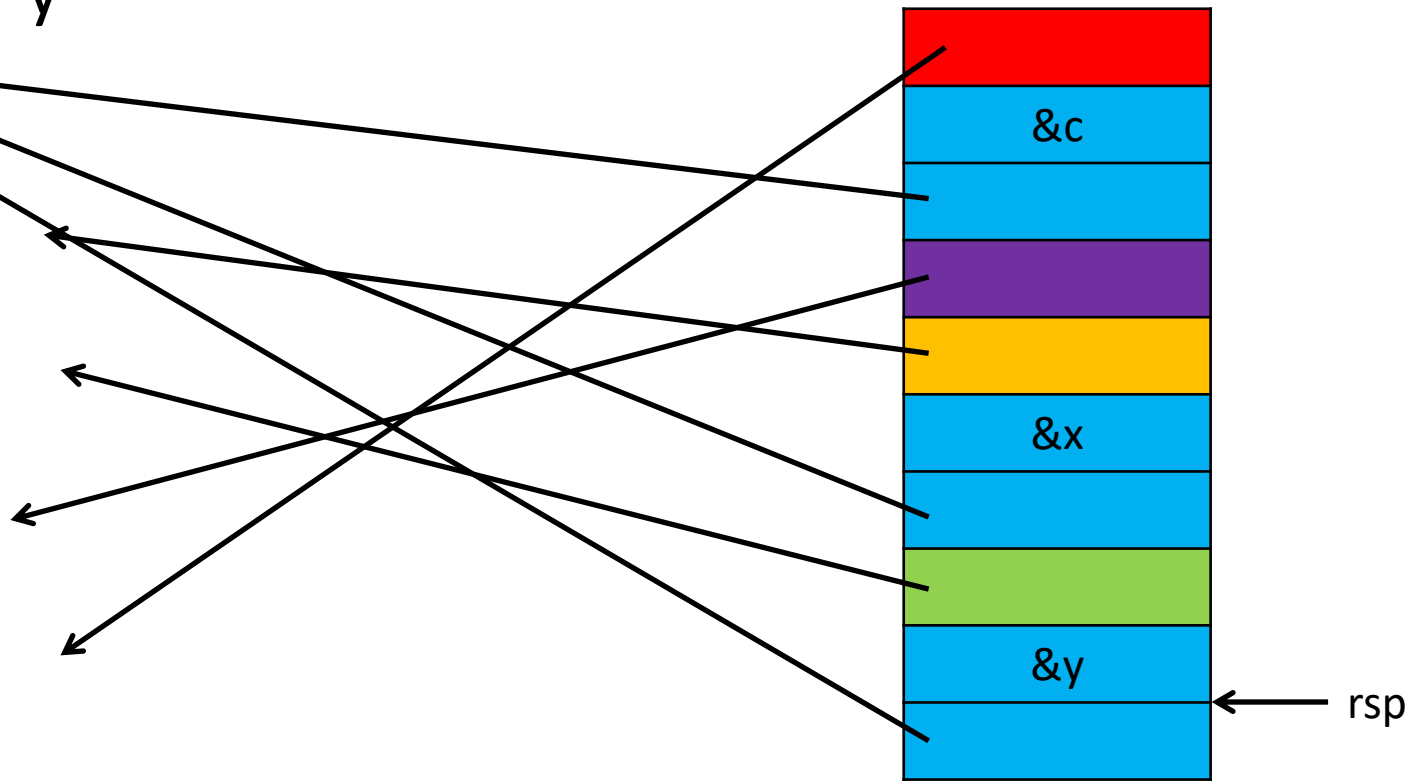

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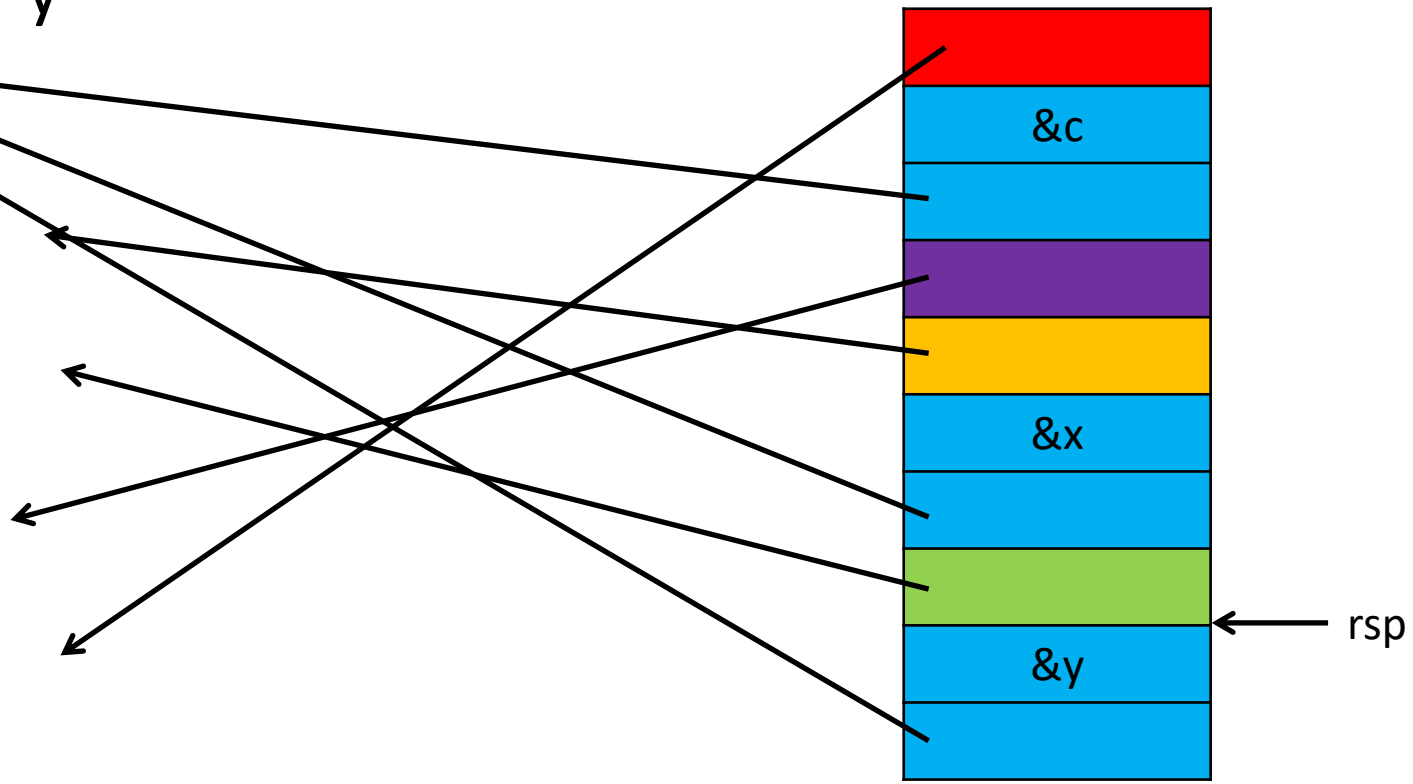

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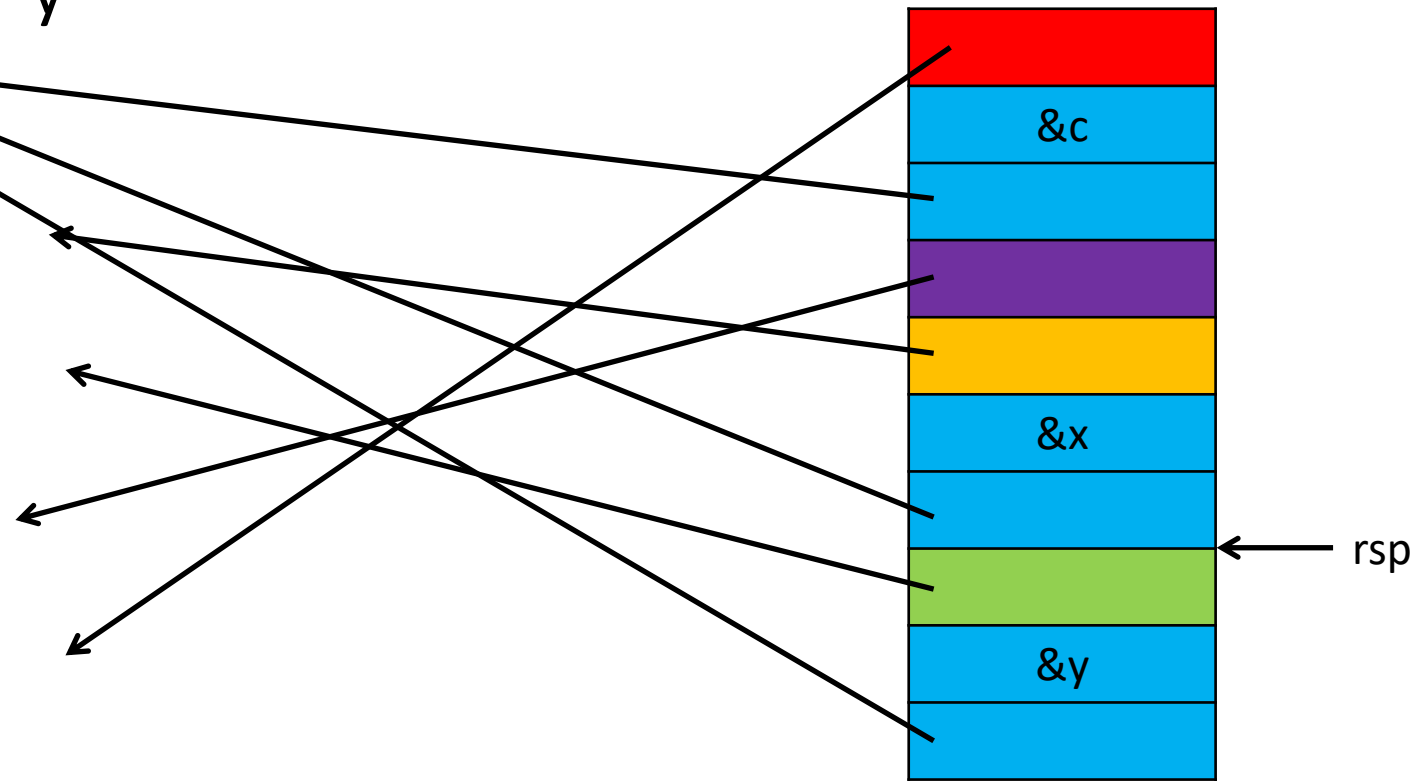

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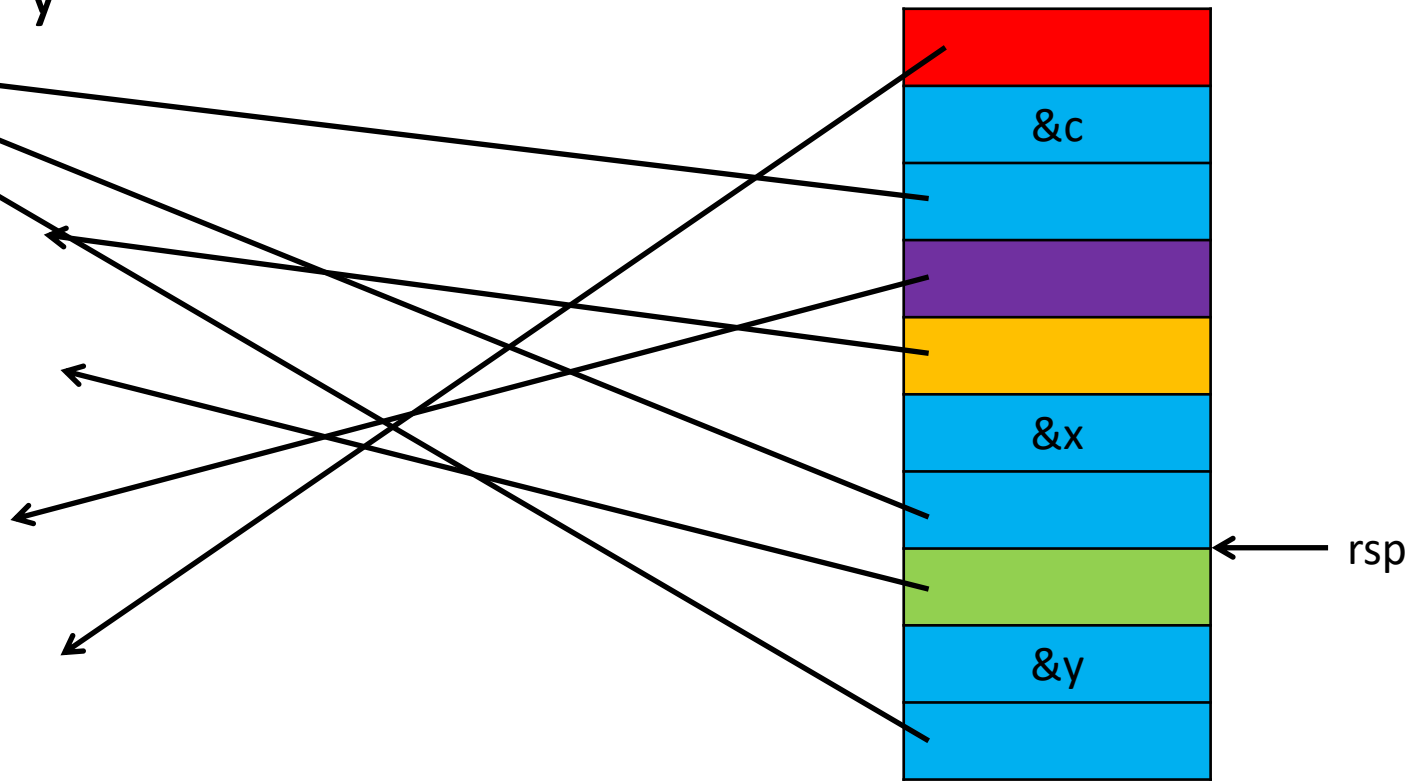

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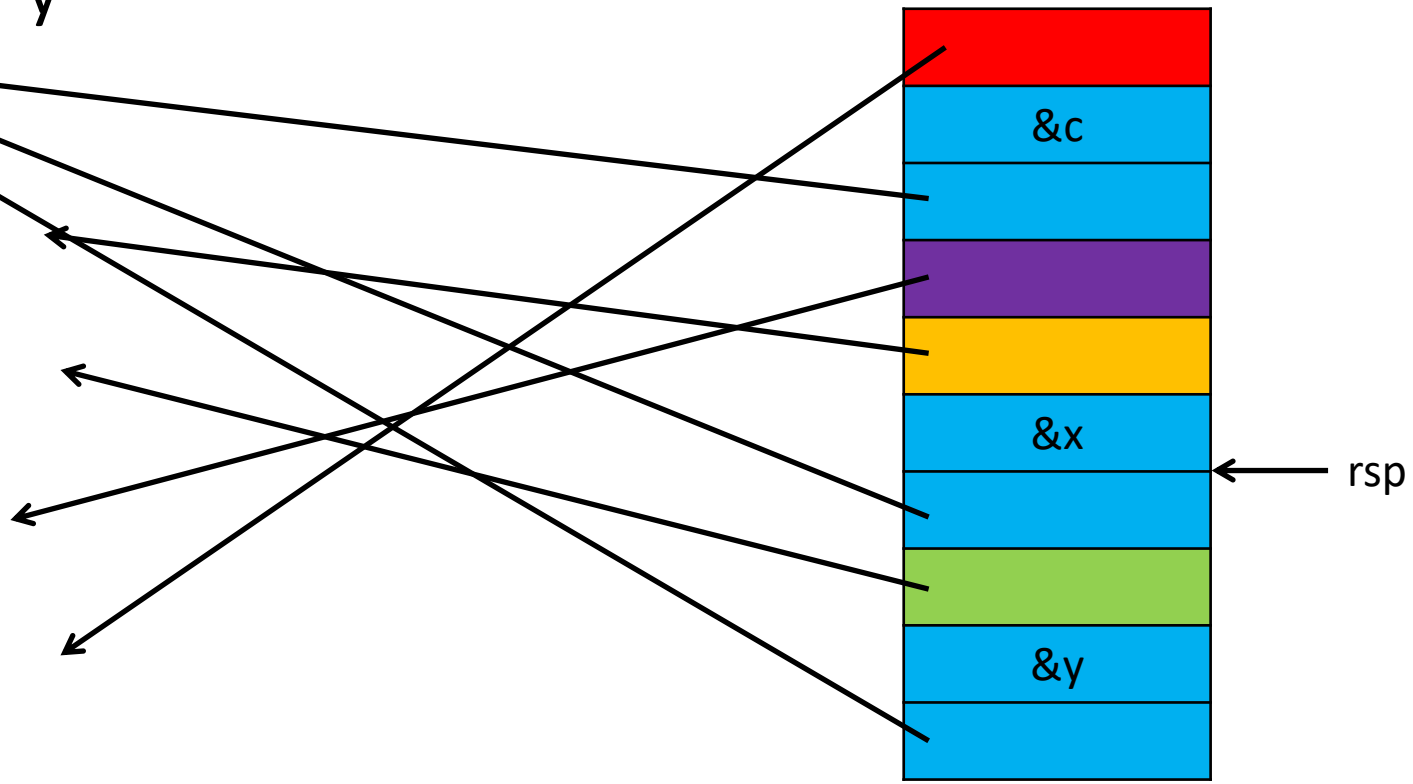

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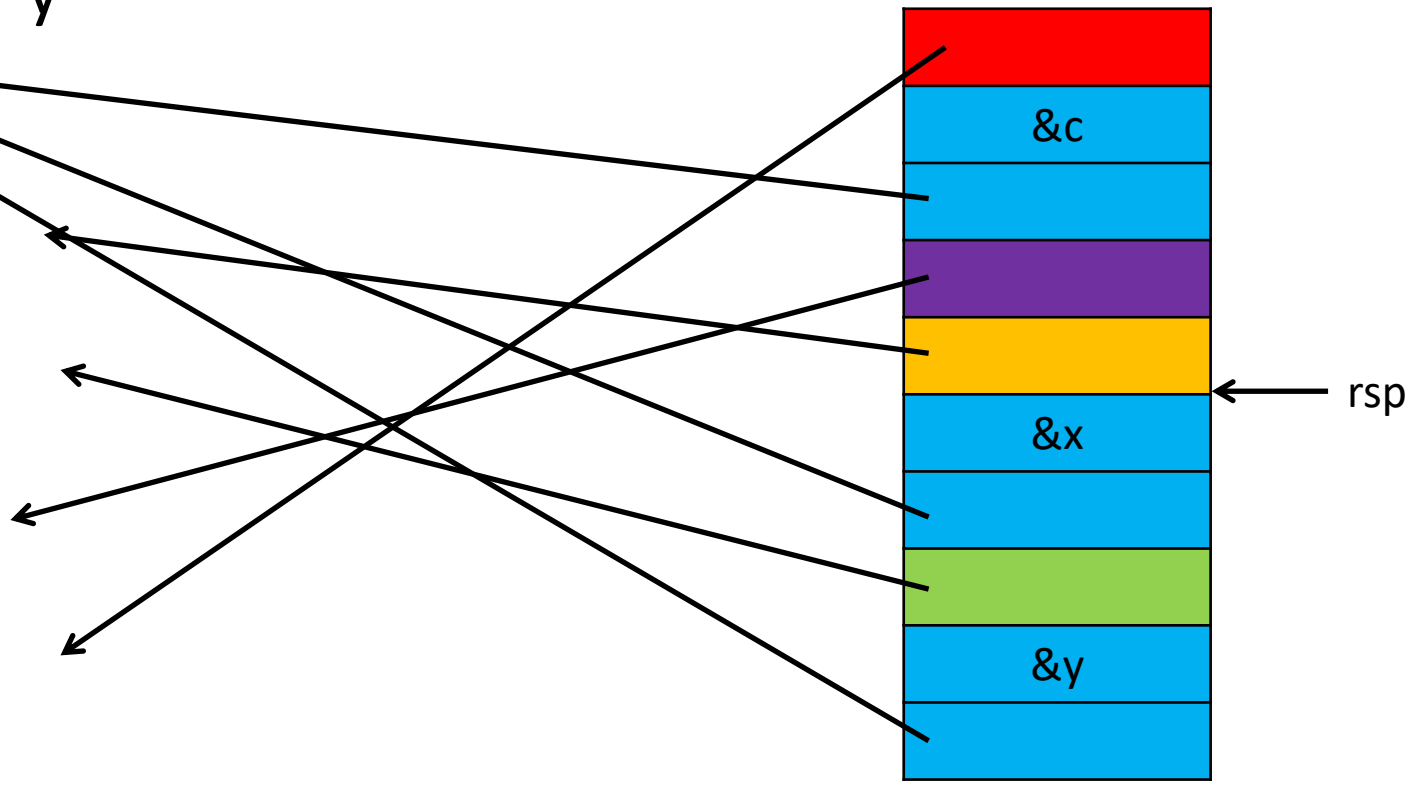

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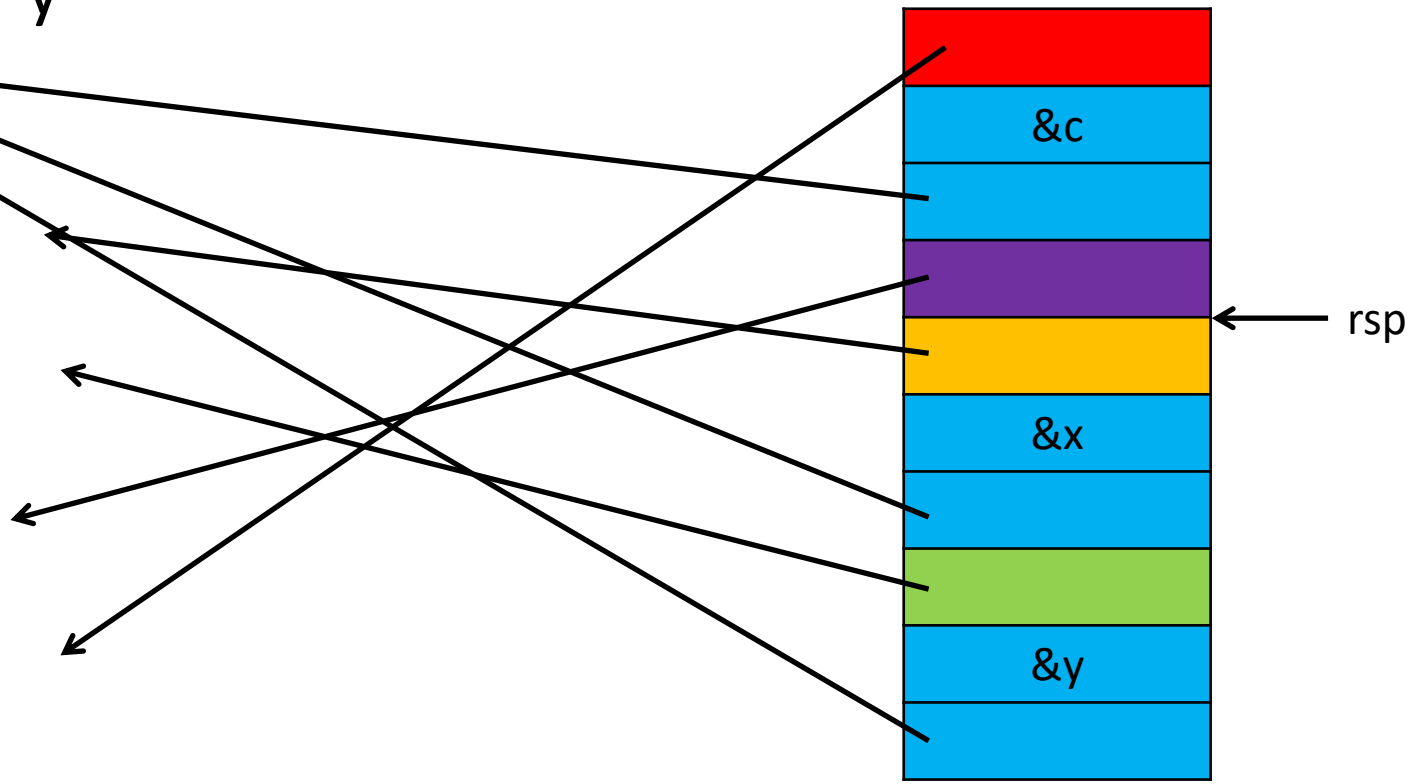

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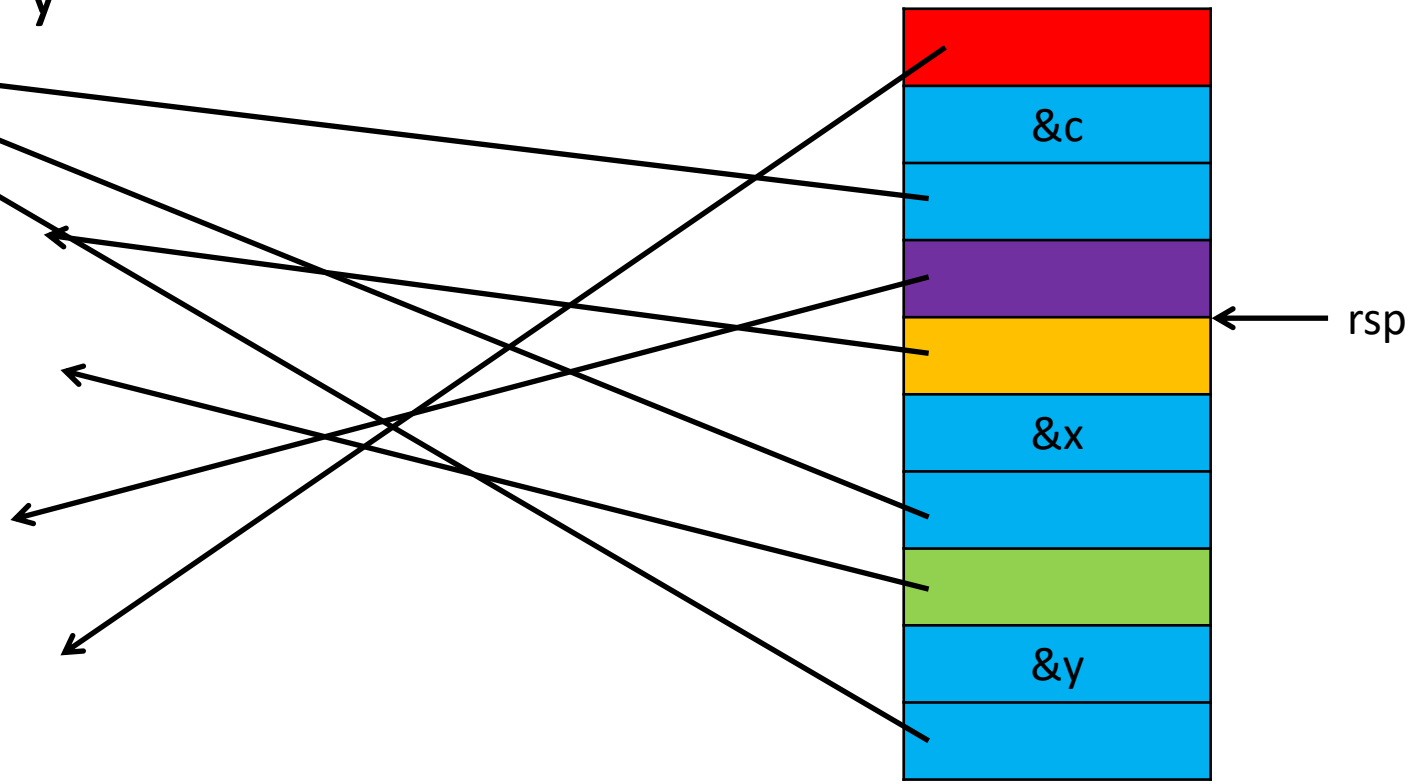

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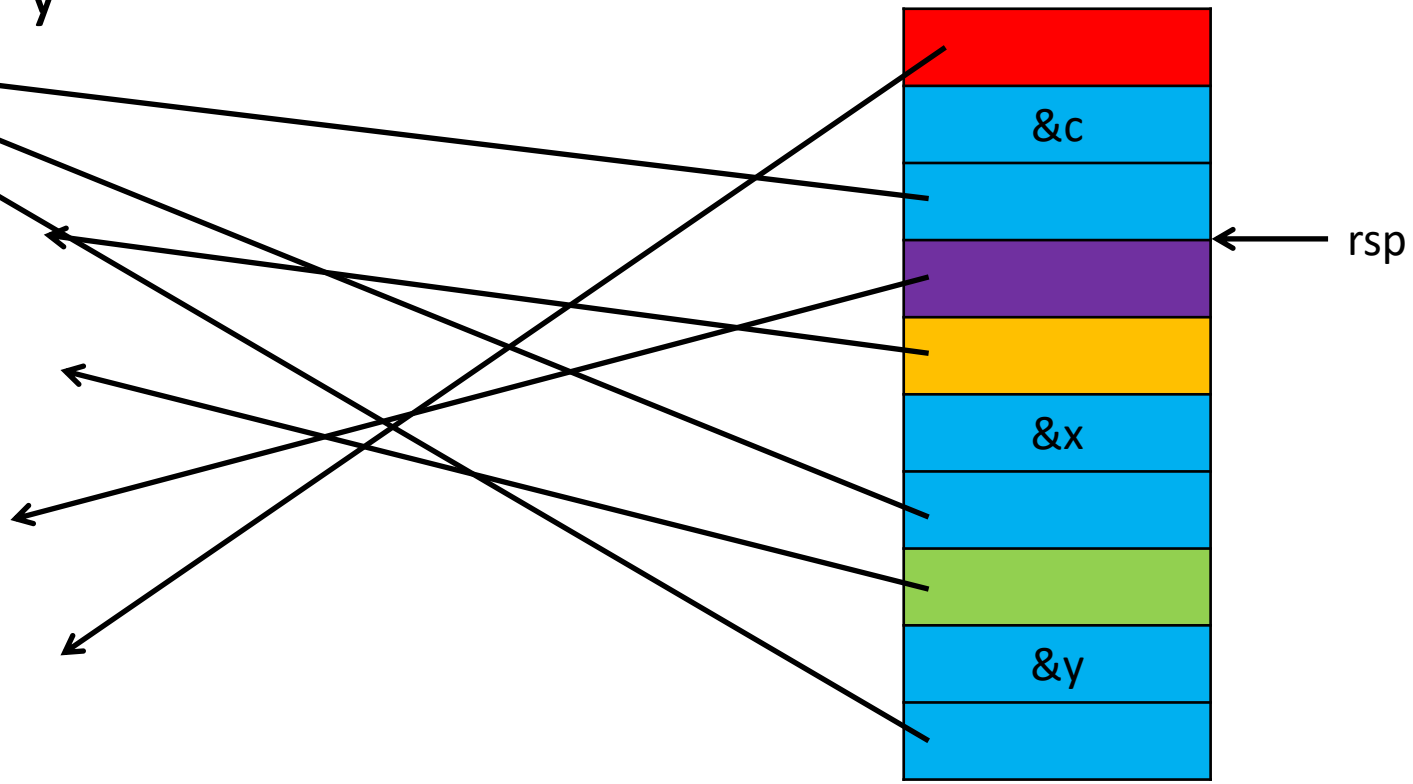

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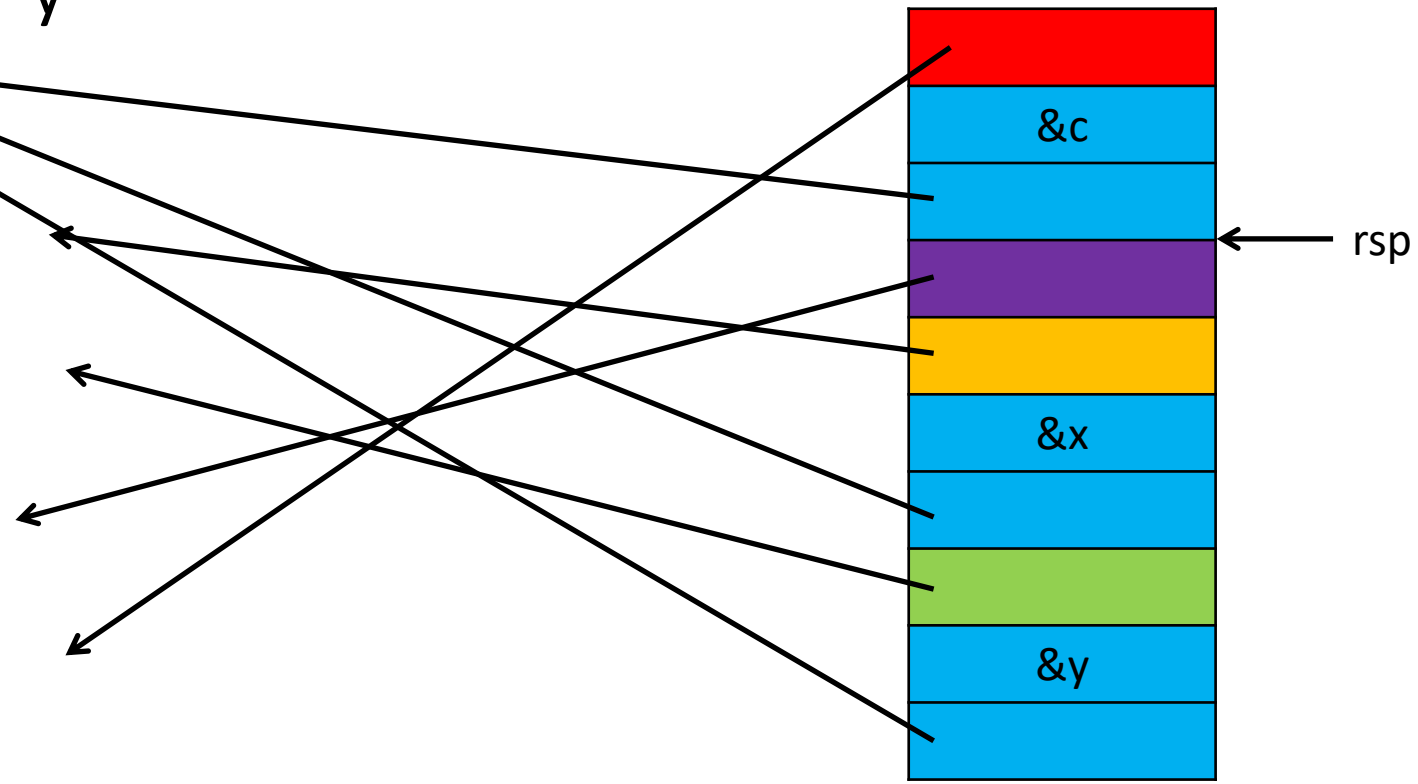

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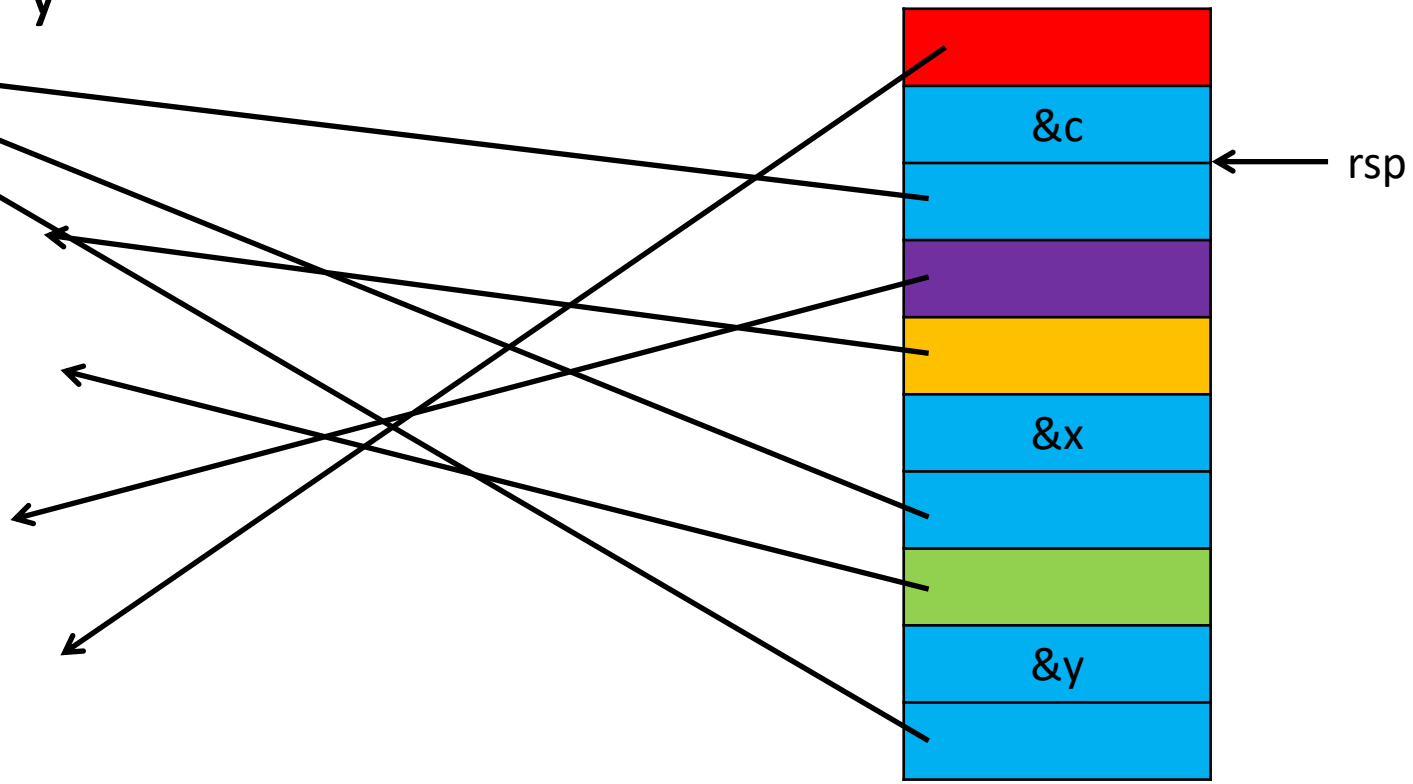

Arithmetic

Register	Value
rax	&c
rbx	y + x

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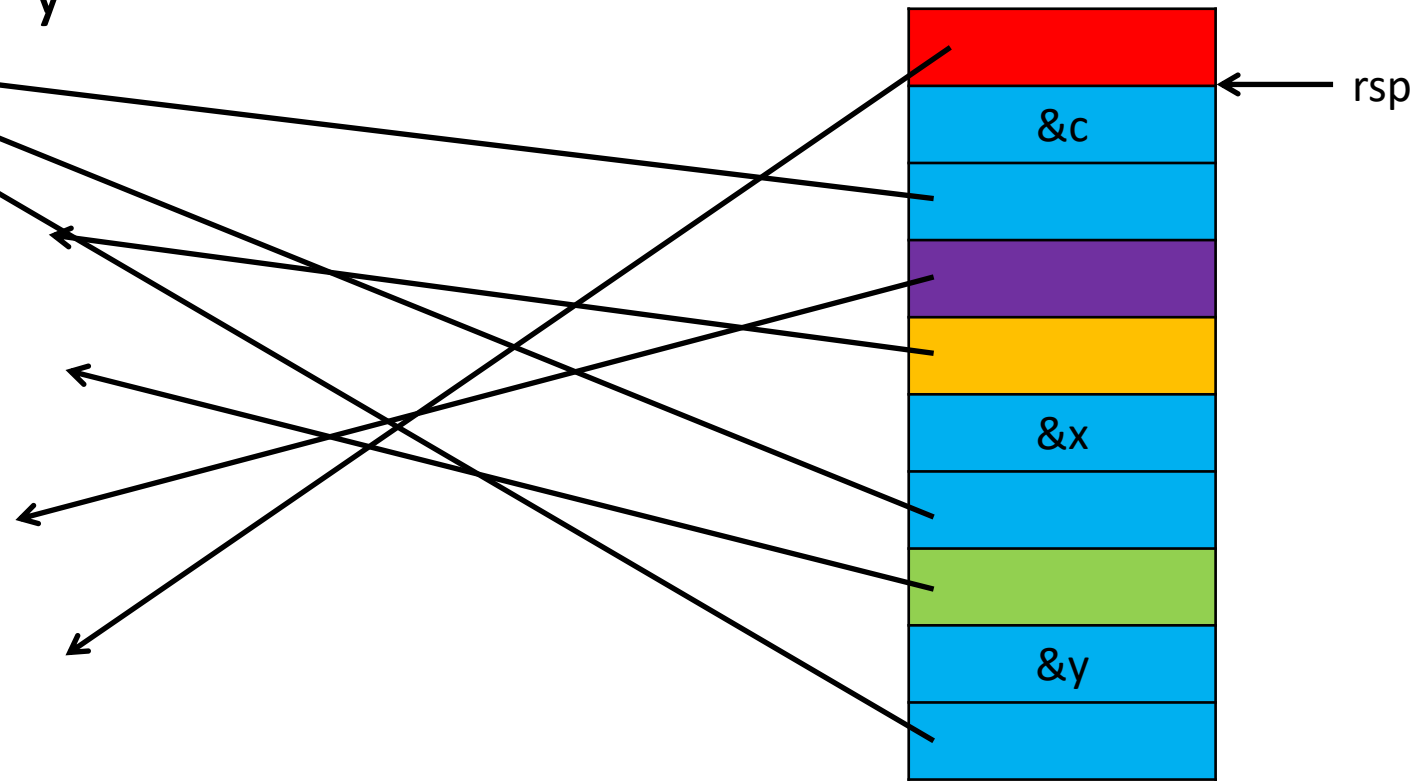

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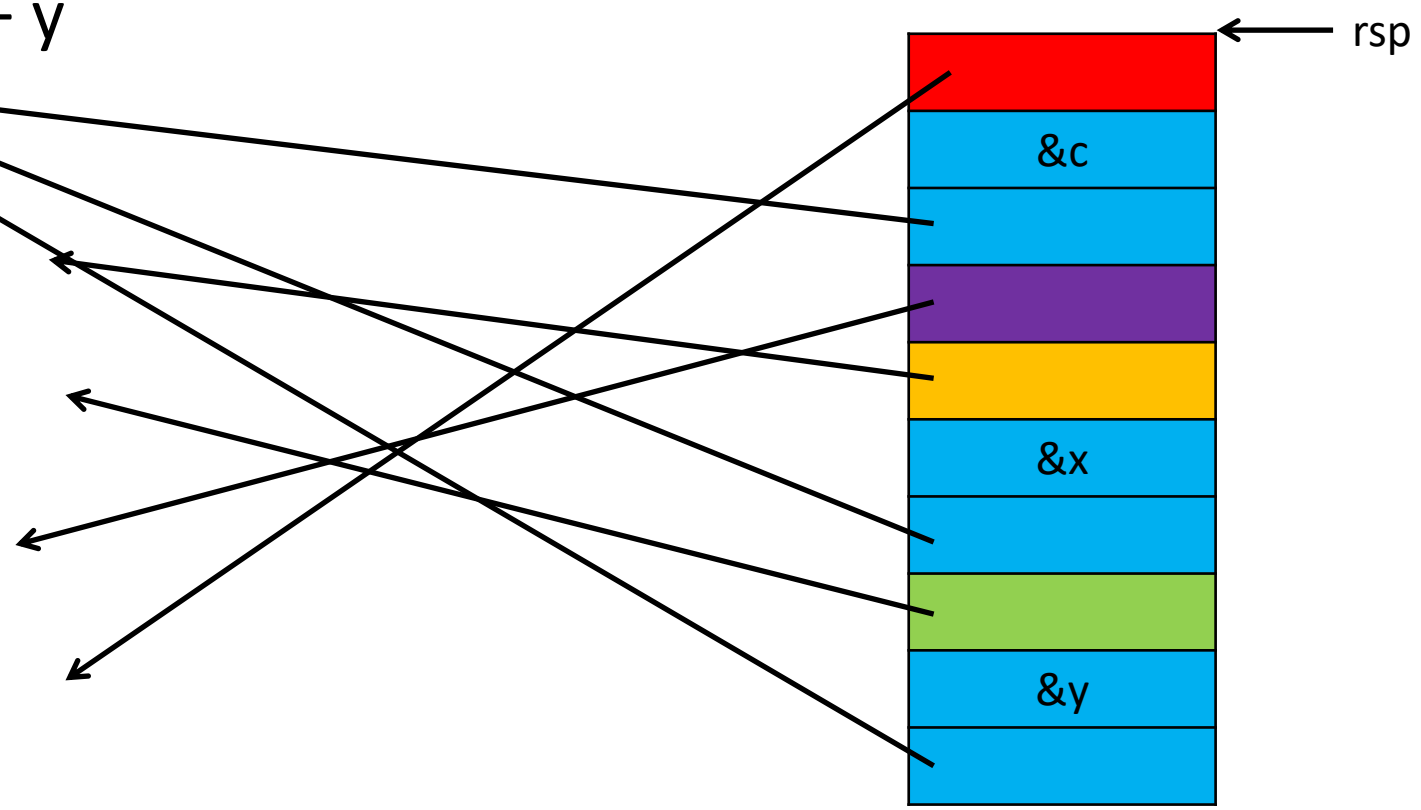

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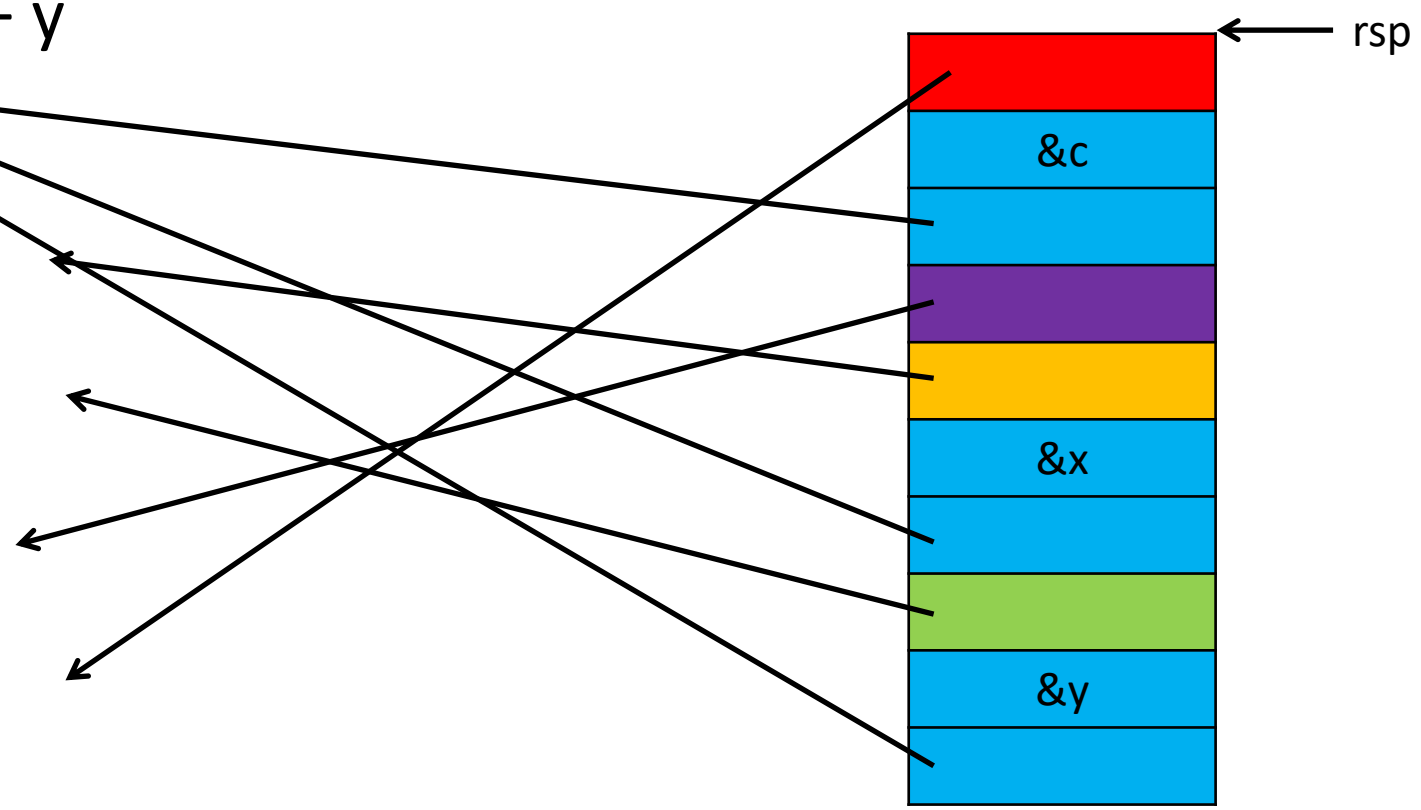

What else can we do?

- Depends on the code we have access to
- Usually: Arbitrary Turing-complete behavior
  - Arithmetic
  - Logic
  - Conditionals and loops
  - Subroutines
  - Calling existing functions
  - System calls
- Sometimes: More limited behavior
  - Often enough for straight-line code and system calls

Comparing ROP to normal programming

	Normal programming	ROP
Instruction pointer	rip	rsp
No-op	nop	ret
Unconditional jump	jmp address	set rsp to address of gadget
Conditional jump	jnz address	set rsp to address of gadget if some condition is met
Variables	memory and registers	mostly memory
Inter-instruction (inter-gadget) register and memory interaction	minimal, mostly explicit; e.g., adding two registers only affects the destination register	can be complex; e.g., adding two registers may involve modifying many registers which impacts other gadgets

Return-oriented conditionals

- Processors support instructions that conditionally change the PC
  - On x86
    - Jcc family: jz, jnz, jl, jle, etc. 33 in total
    - loop, loope, loopne
    - Based on condition codes mostly; and on rcx for some
  - On MIPS
    - beq and bne
    - Based on comparison of registers
- Processors generally don’t support for conditionally changing the stack pointer (with some exceptions)

We want conditional jumps

- Unconditional jump addr
  - pop rax
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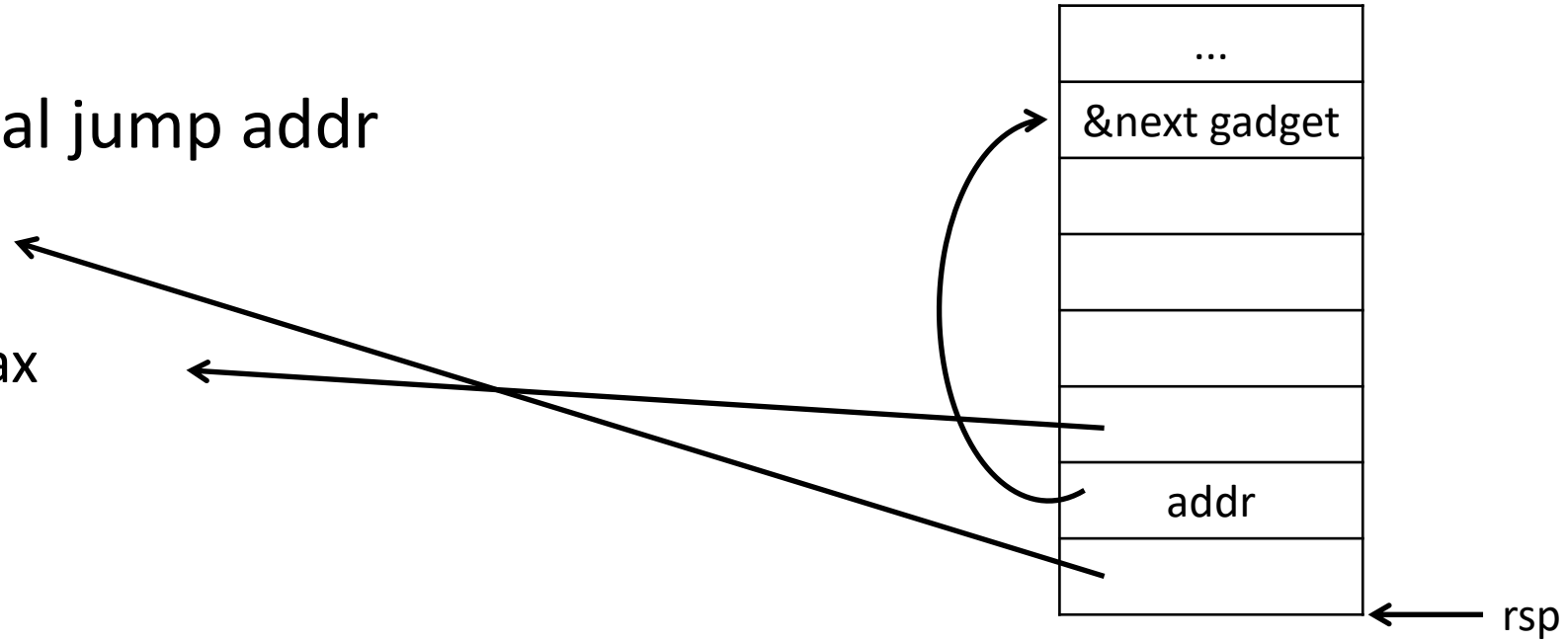

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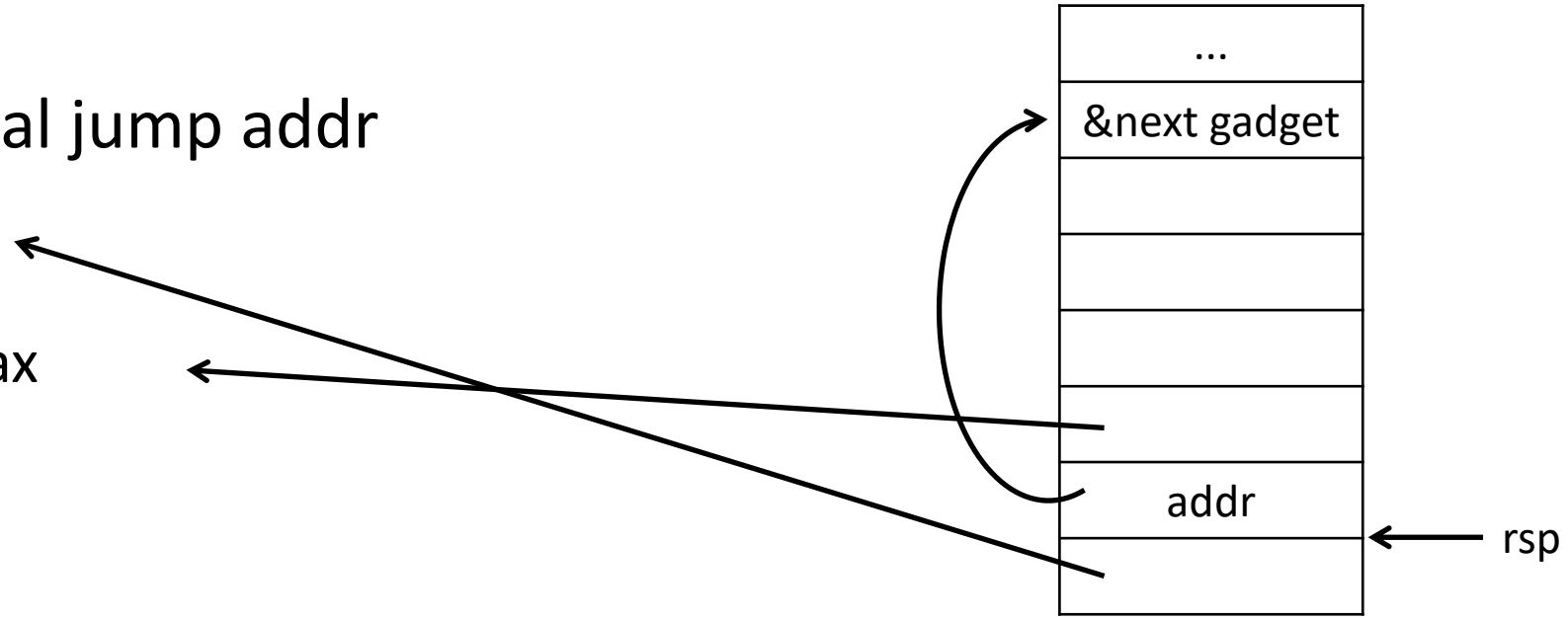

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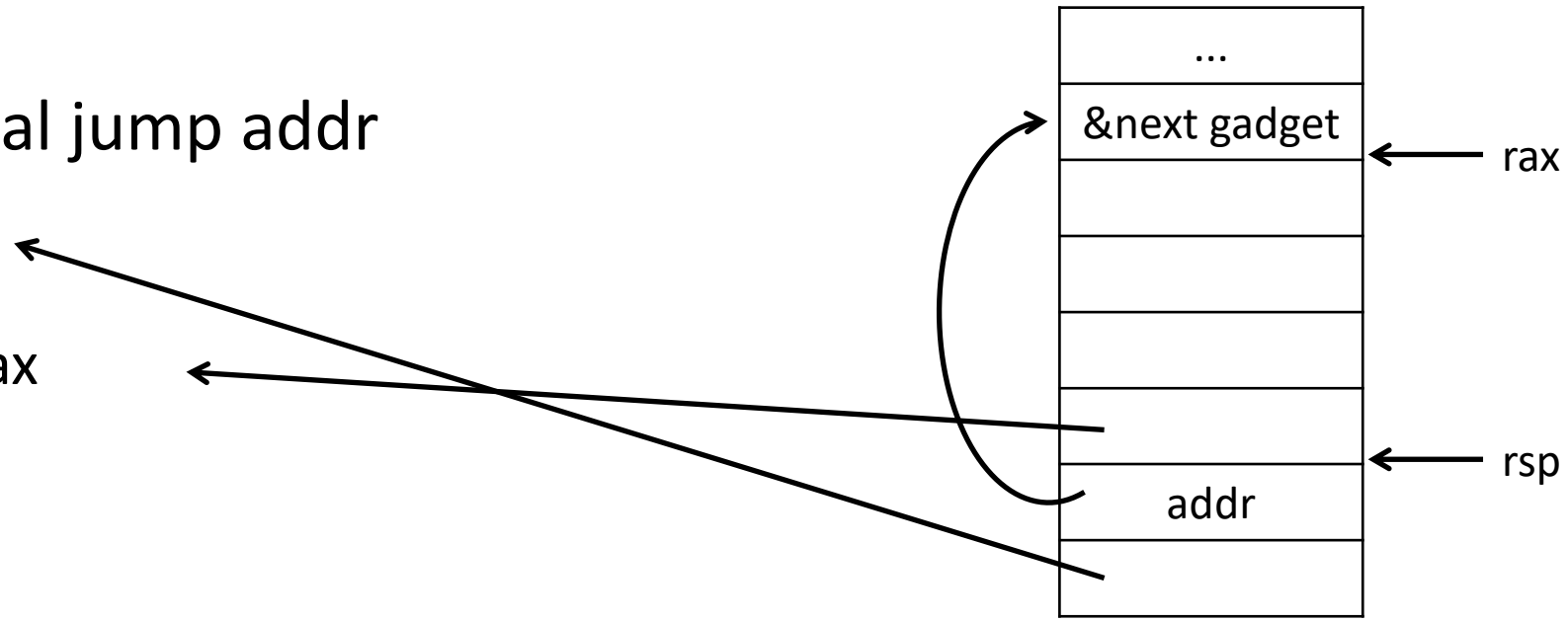

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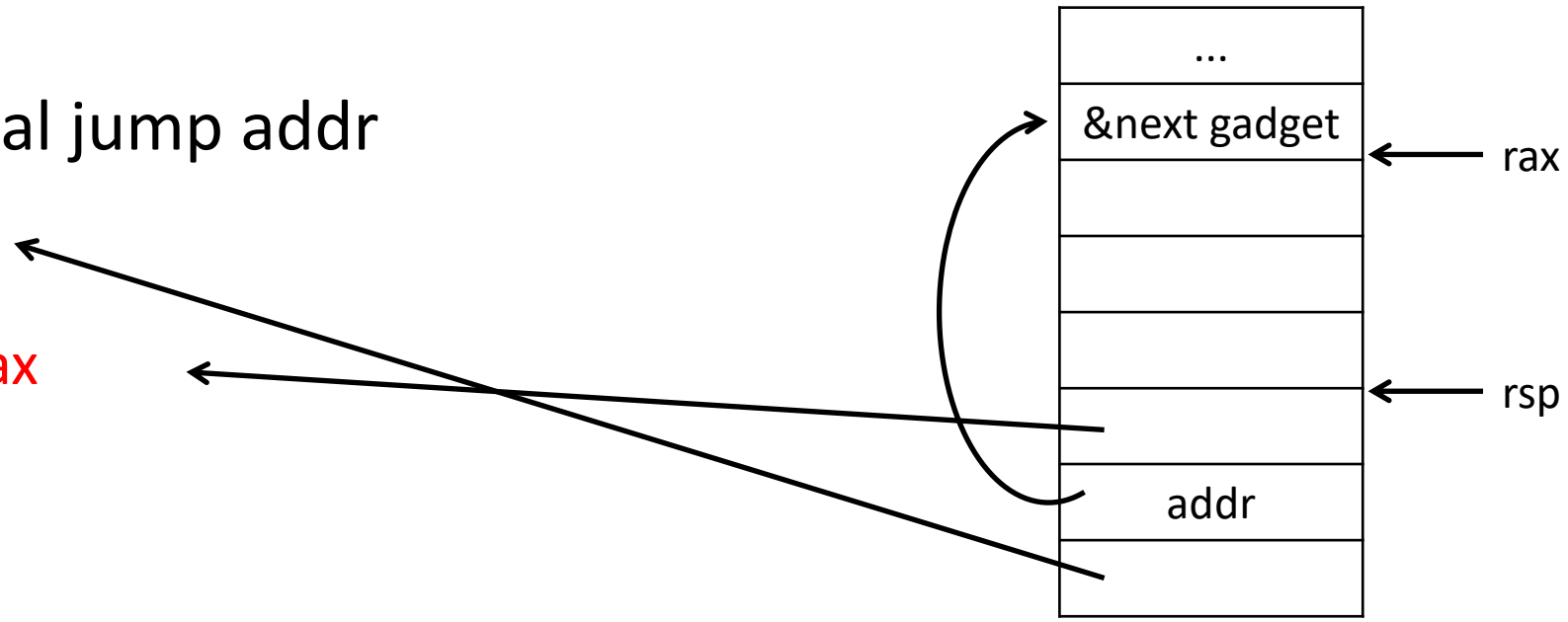

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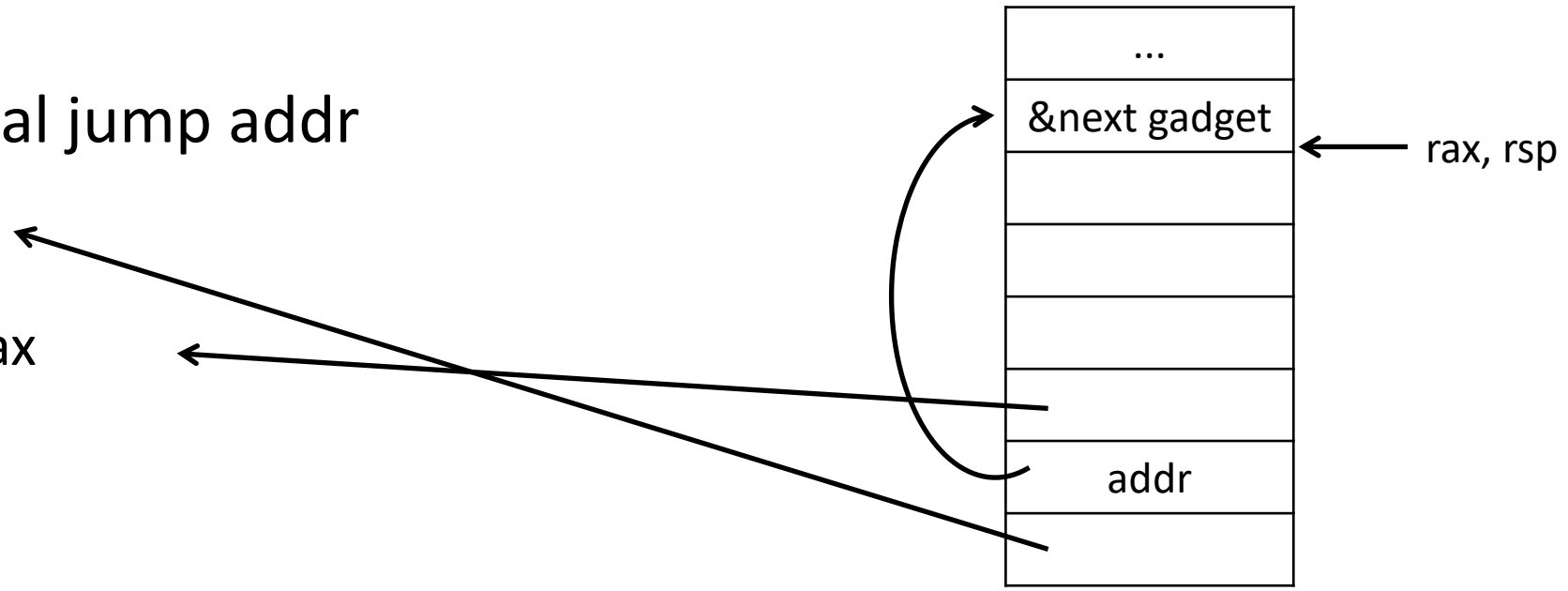

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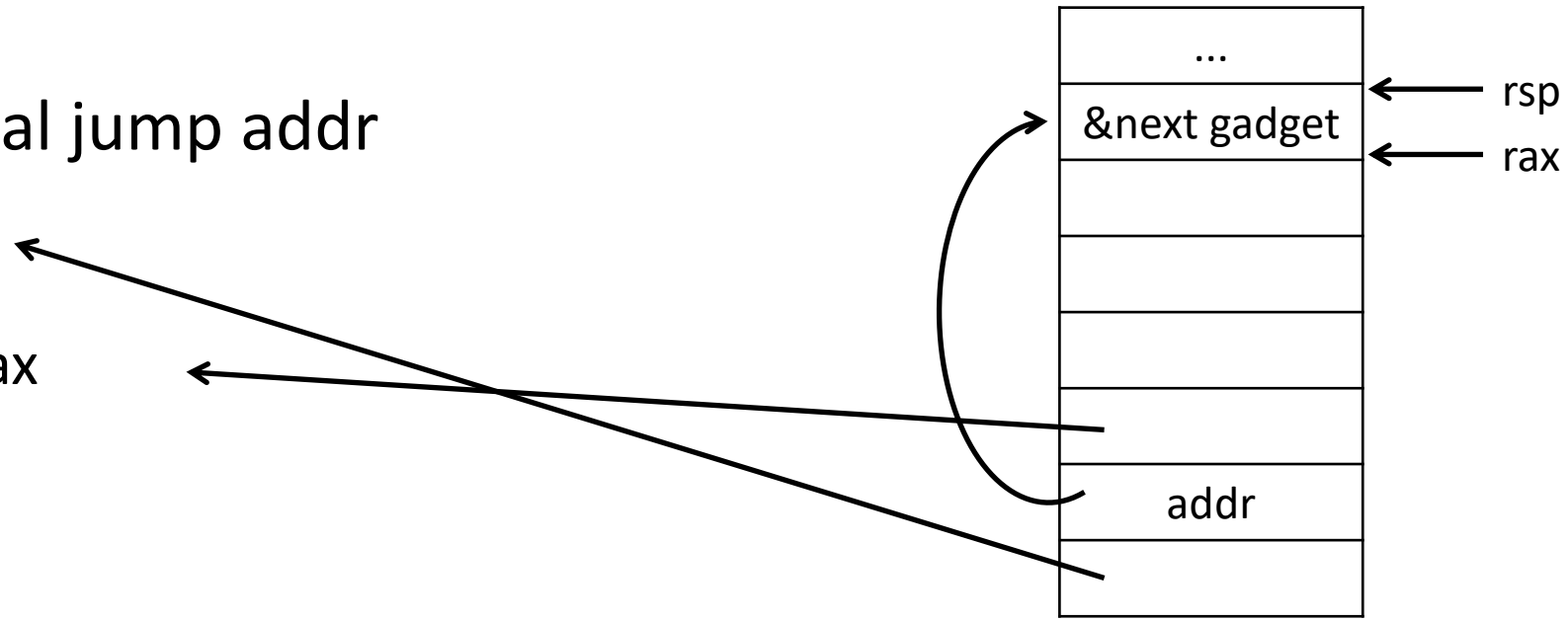


We want conditional jumps

- Unconditional jump addr
  - pop rax
    ret
  - mov rsp, rax
    ret
- Conditional jump addr, one way
  - Conditionally set a register to 0 or 0xffffffffffffffff = -1
  - Perform a logical AND with the register and an offset
  - Add the result to rsp
- Another approach: use conditional move instructions cmovz, cmovnz after setting the zero flag via cmp or test

Conditionally set a register to 0 or -1

- Compare registers rax and rbx and set rcx to
  - -1 if rax < rbx
  - 0 if rax >= rbx
- Ideally we would find a sequence like

```
cmp rax, rbx        set carry flag cf according to rax - rbx
sbb rcx, rcx        rcx ← rcx - rcx - cf; or rcx ← -cf
ret
```

- Unlikely to find this; instead look for cmp; ret and sbb; ret sequences

Performing a logical AND with a constant

- Pop the constant into a register using pop; ret
- Use an and; ret sequence

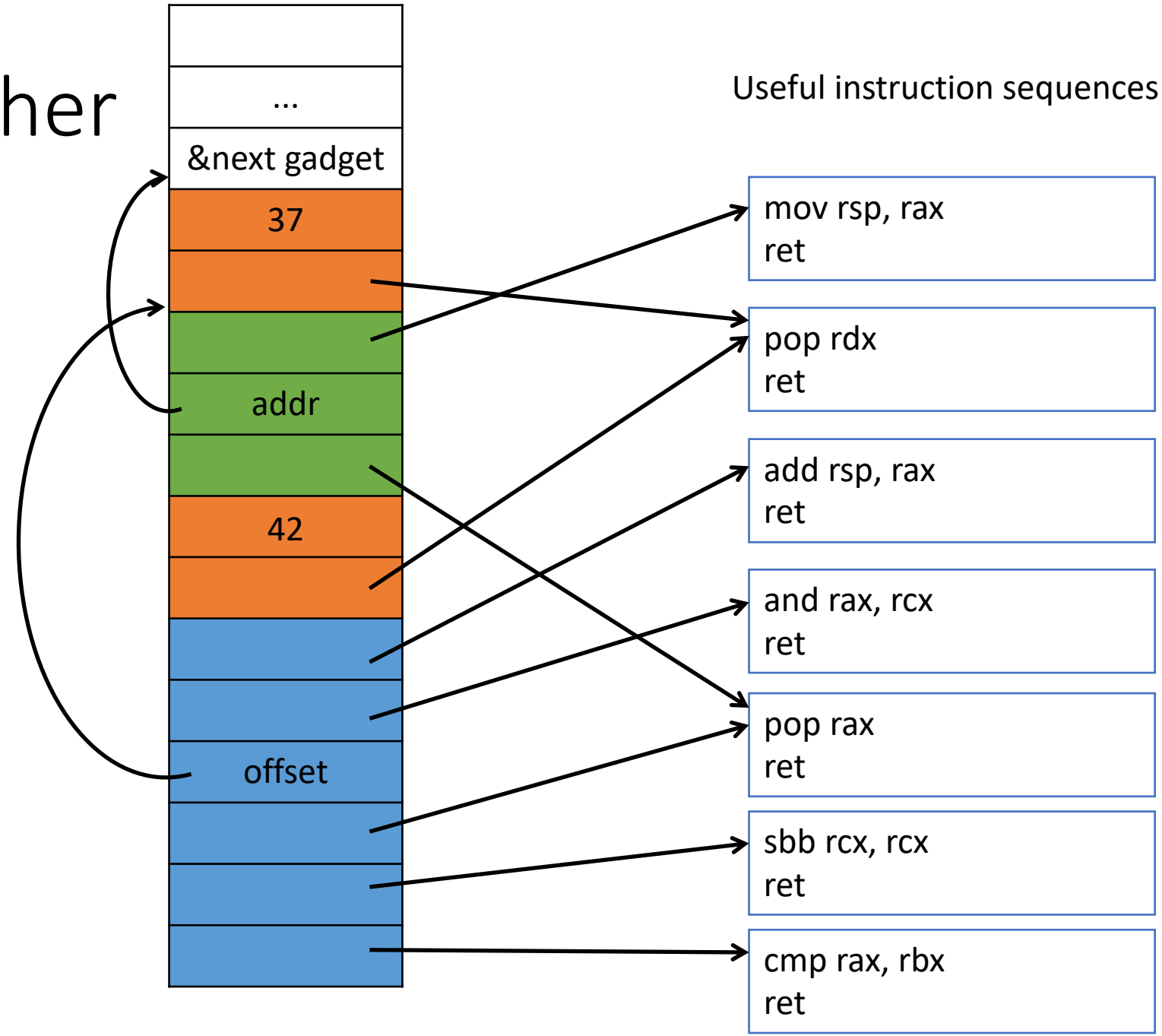
Updating the stack pointer

- Use an add rsp, reg; ret sequence

- Other options available, e.g., if rsp + offset is in rax
  - push rax ; pop rsp ; ret (has the downside that it modified the stack which is an issue for loops)
  - xchg rax, rsp

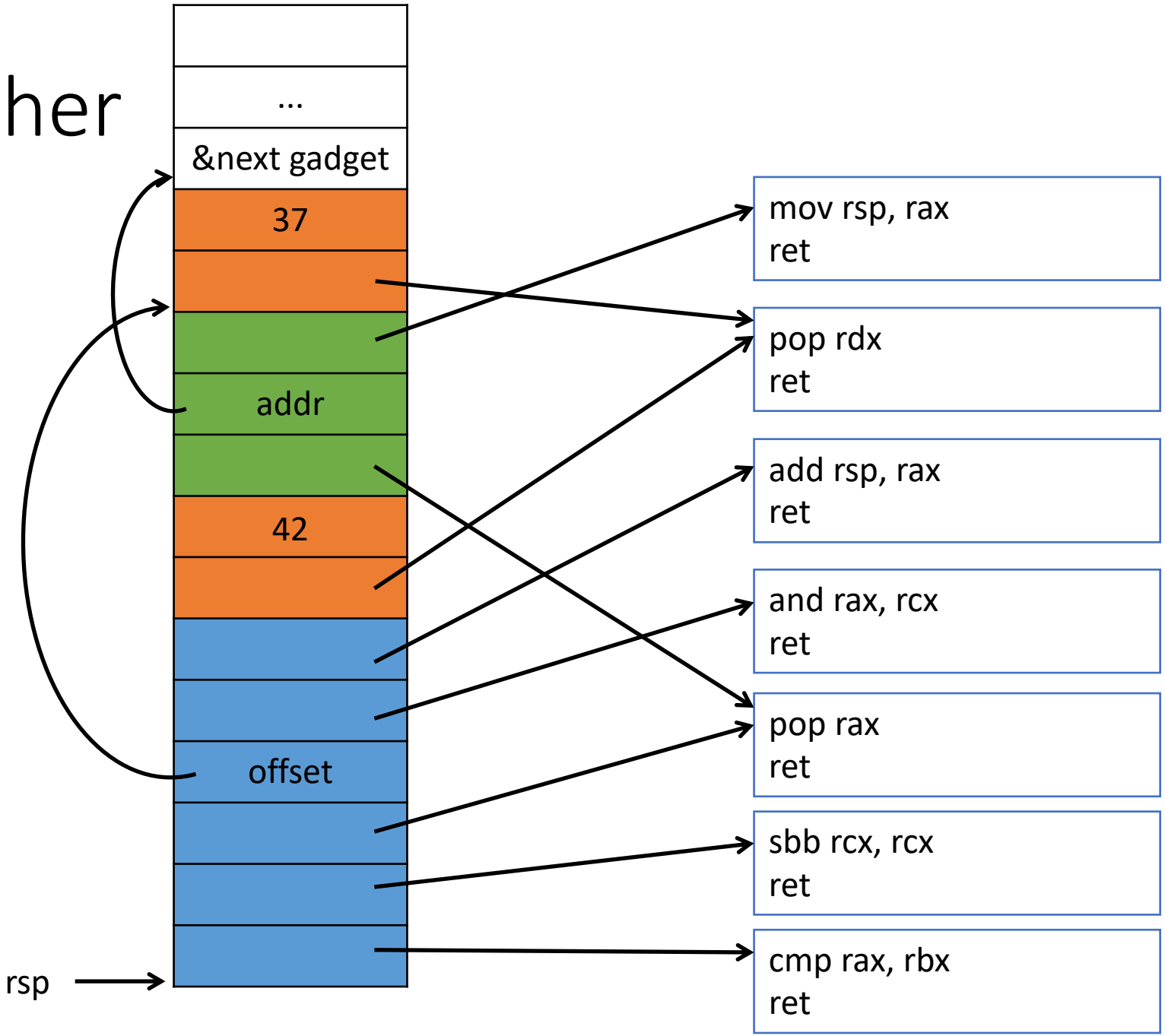
Putting it together

Conditional jump
Load constant in rdx
Unconditional jump



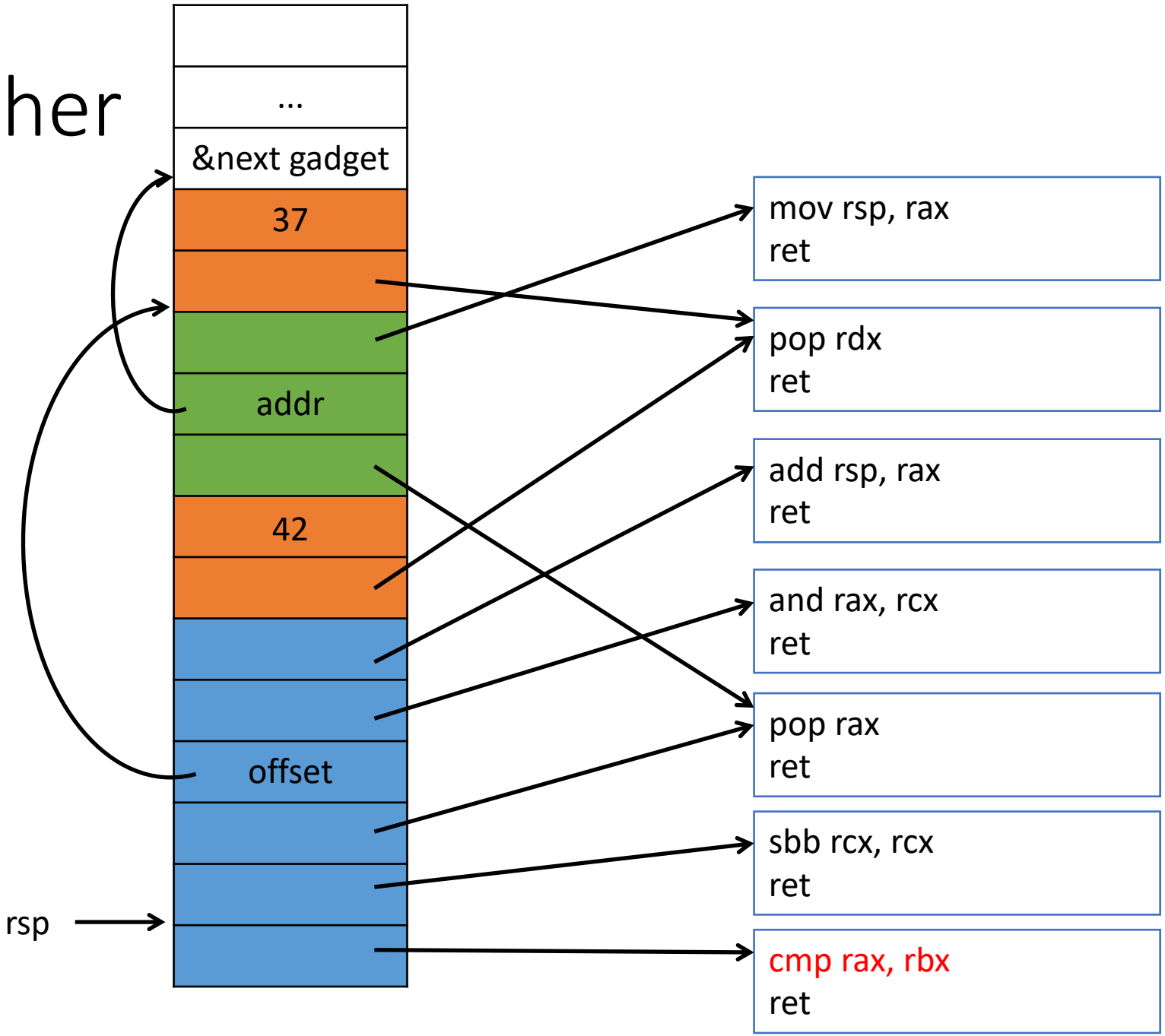
Putting it together

Register	Value
rax	10
rbx	20
rcx	108
rdx	17



Putting it together

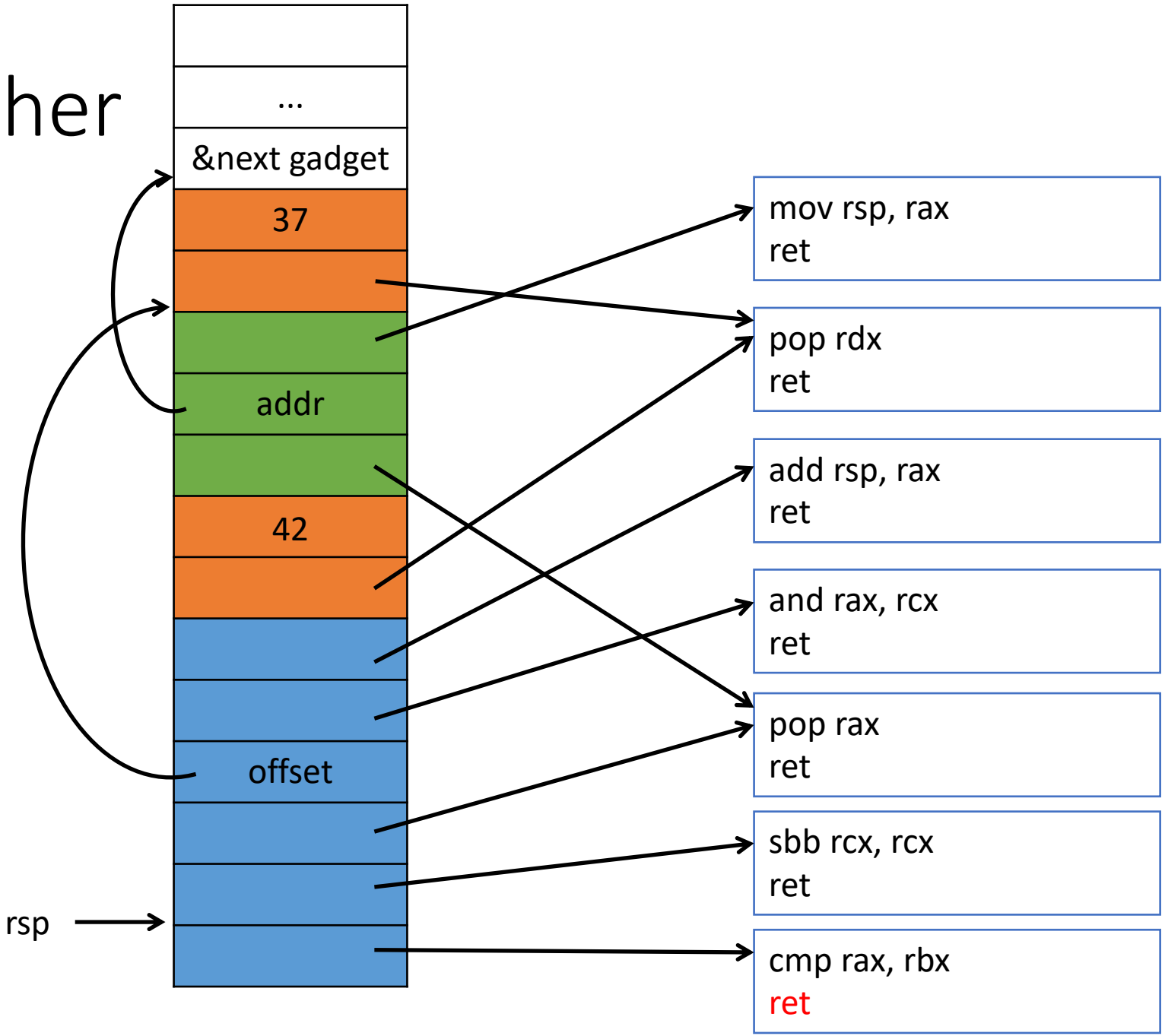
Register	Value
rax	10
rbx	20
rcx	108
rdx	17



Putting it together

Register	Value
rax	10
rbx	20
rcx	108
rdx	17

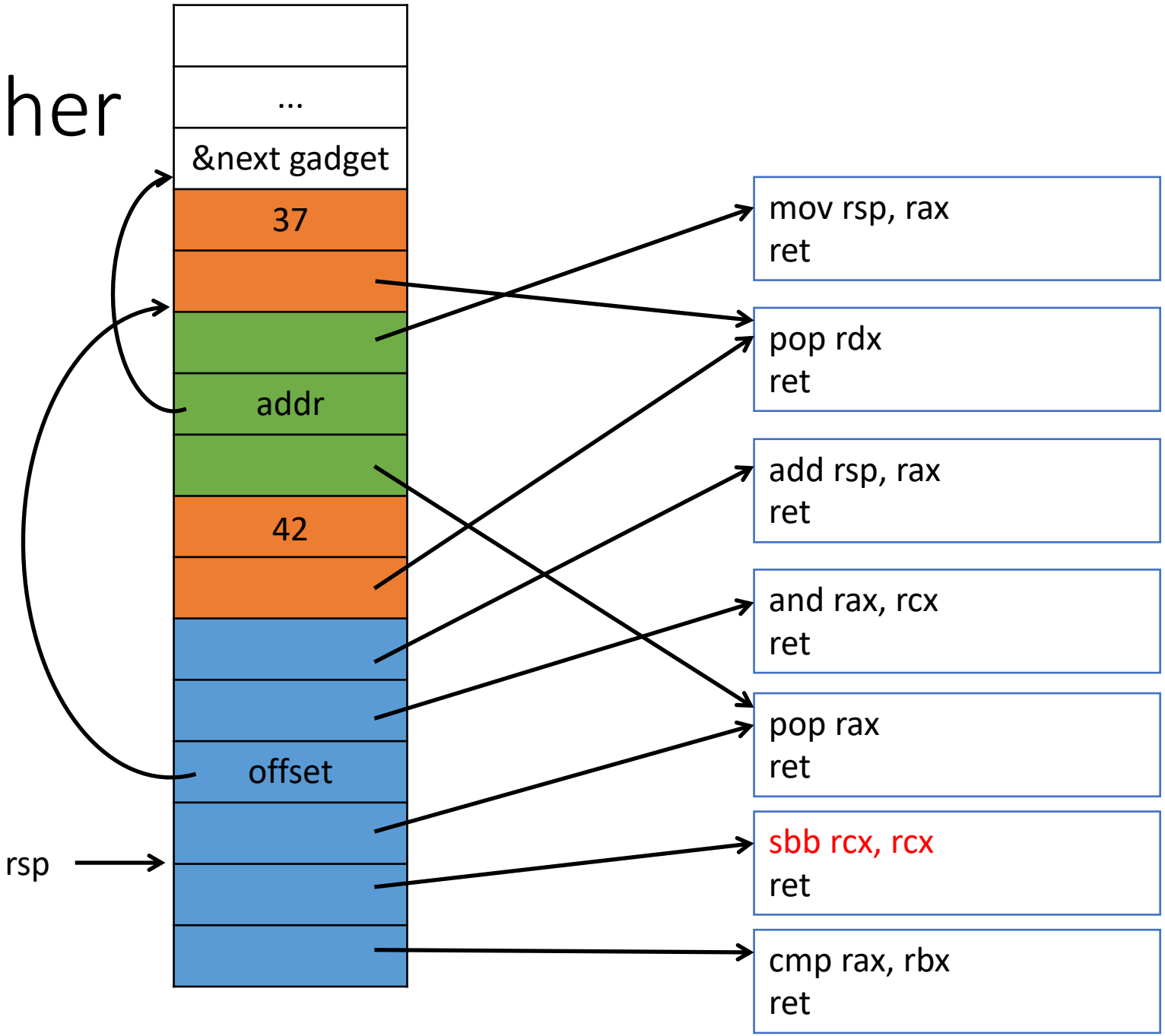
cf = 1



Putting it together

Register	Value
rax	10
rbx	20
rcx	108
rdx	17

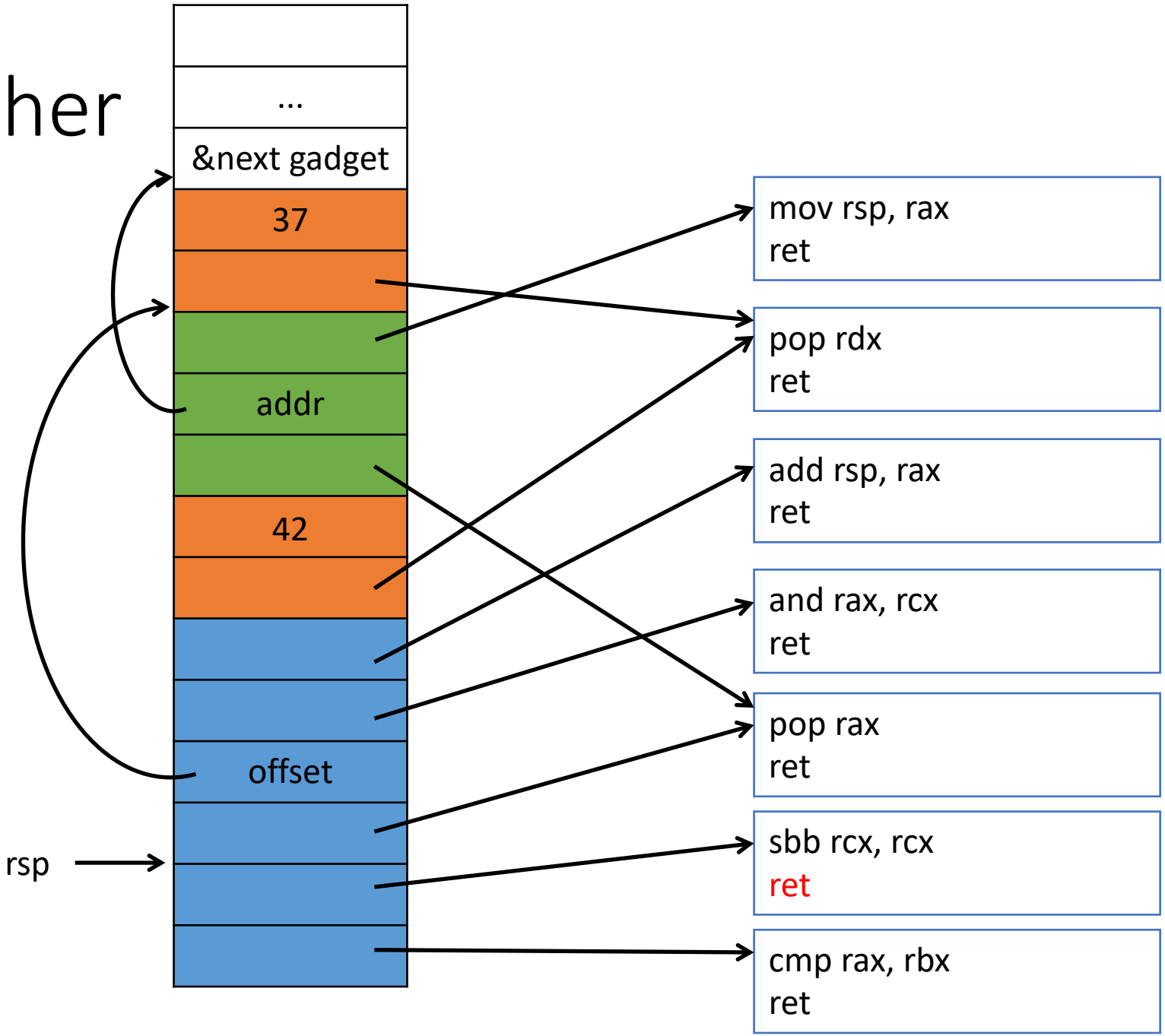
cf = 1



Putting it together

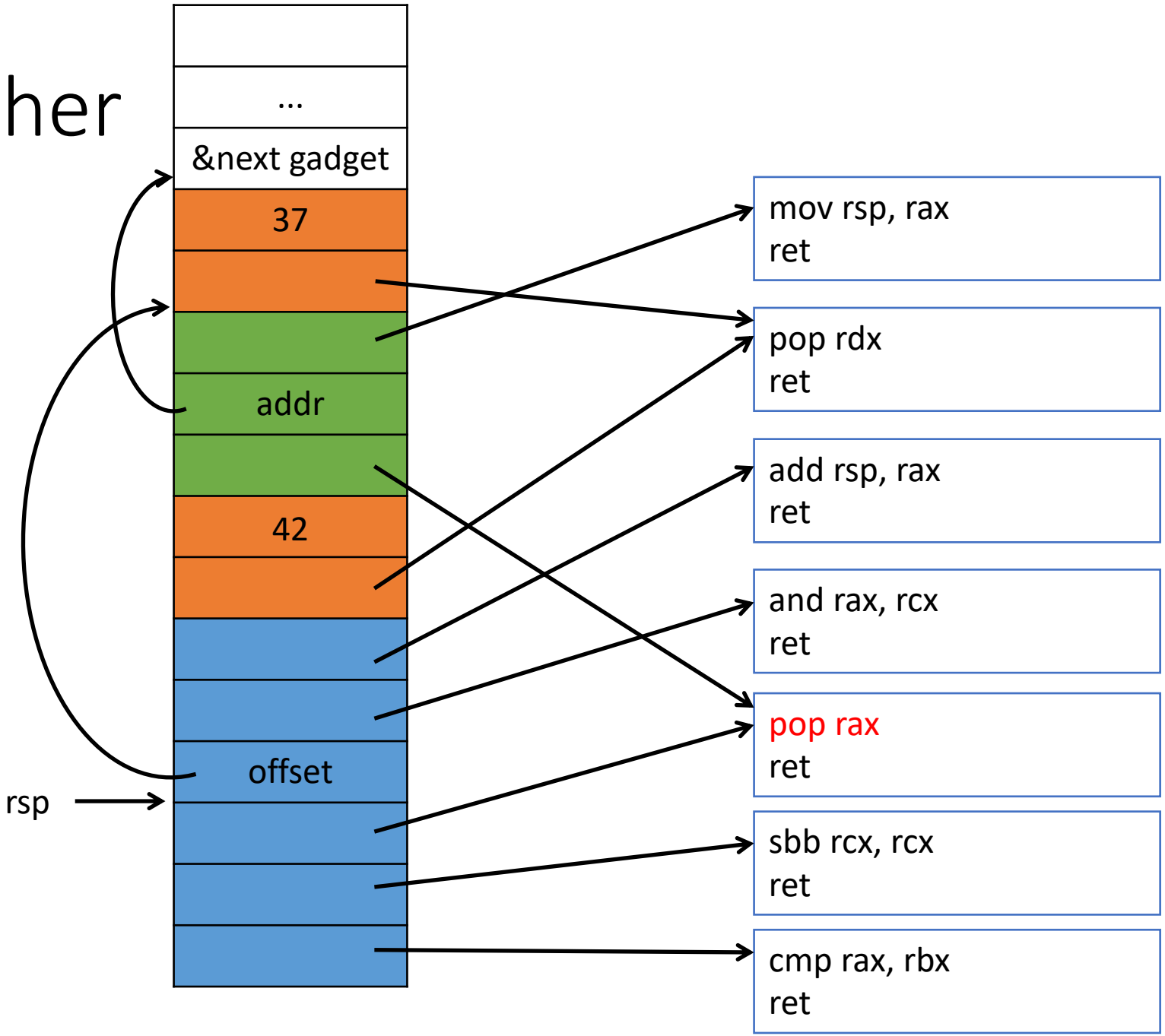
Register	Value
rax	10
rbx	20
rcx	-1
rdx	17

cf = 1



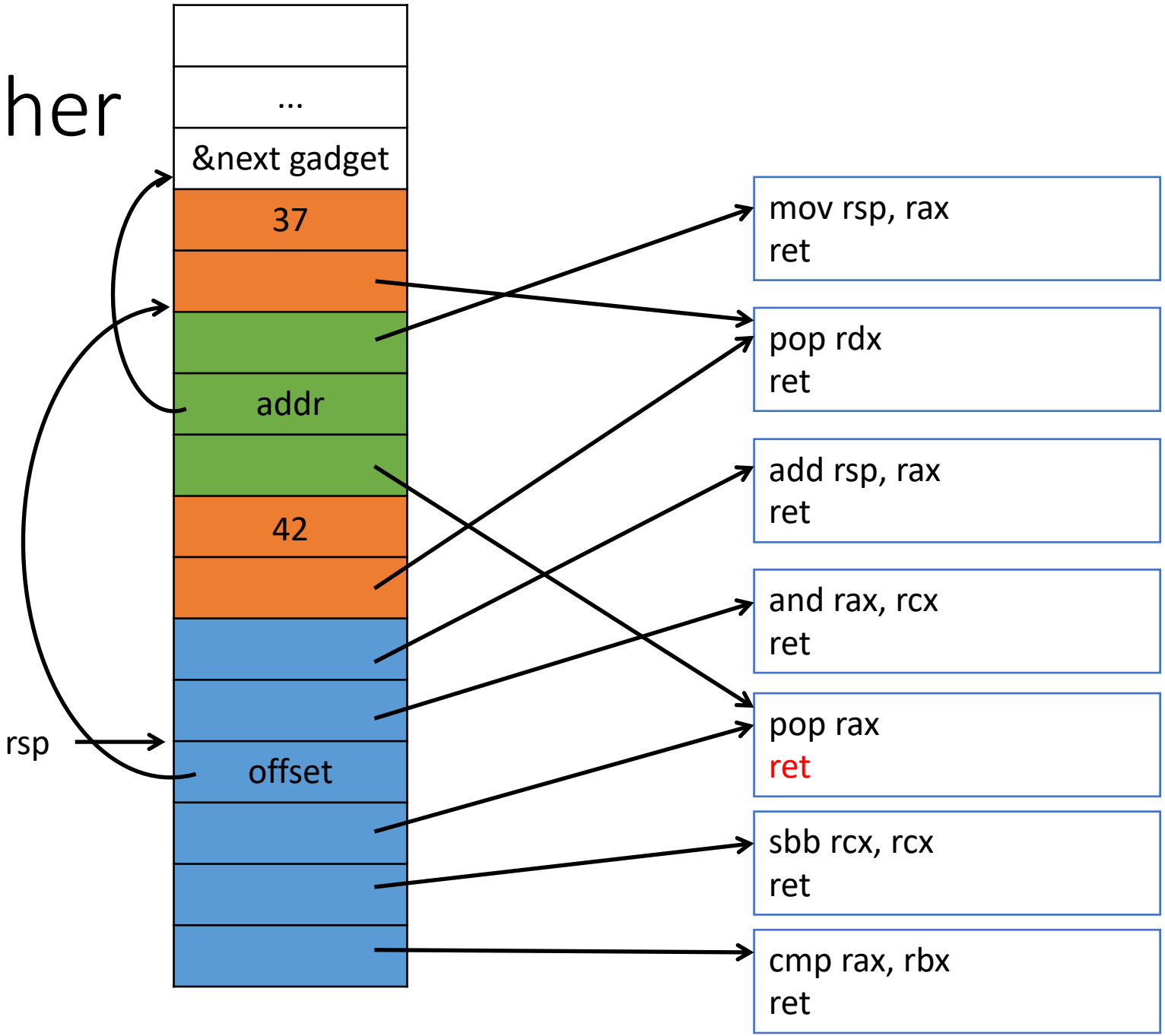
Putting it together

Register	Value
rax	10
rbx	20
rcx	-1
rdx	17



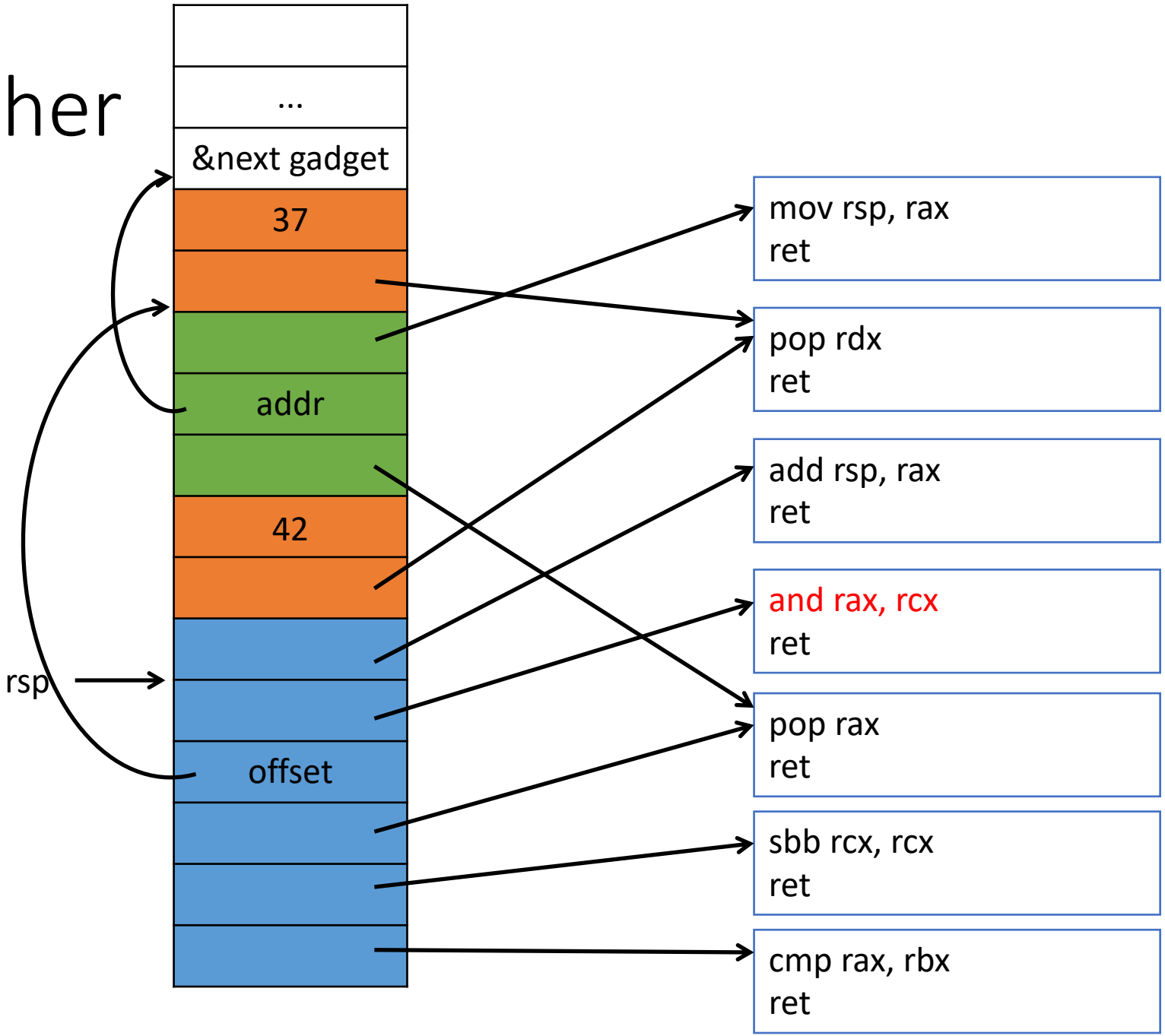
Putting it together

Register	Value
rax	40 = offset
rbx	20
rcx	-1
rdx	17



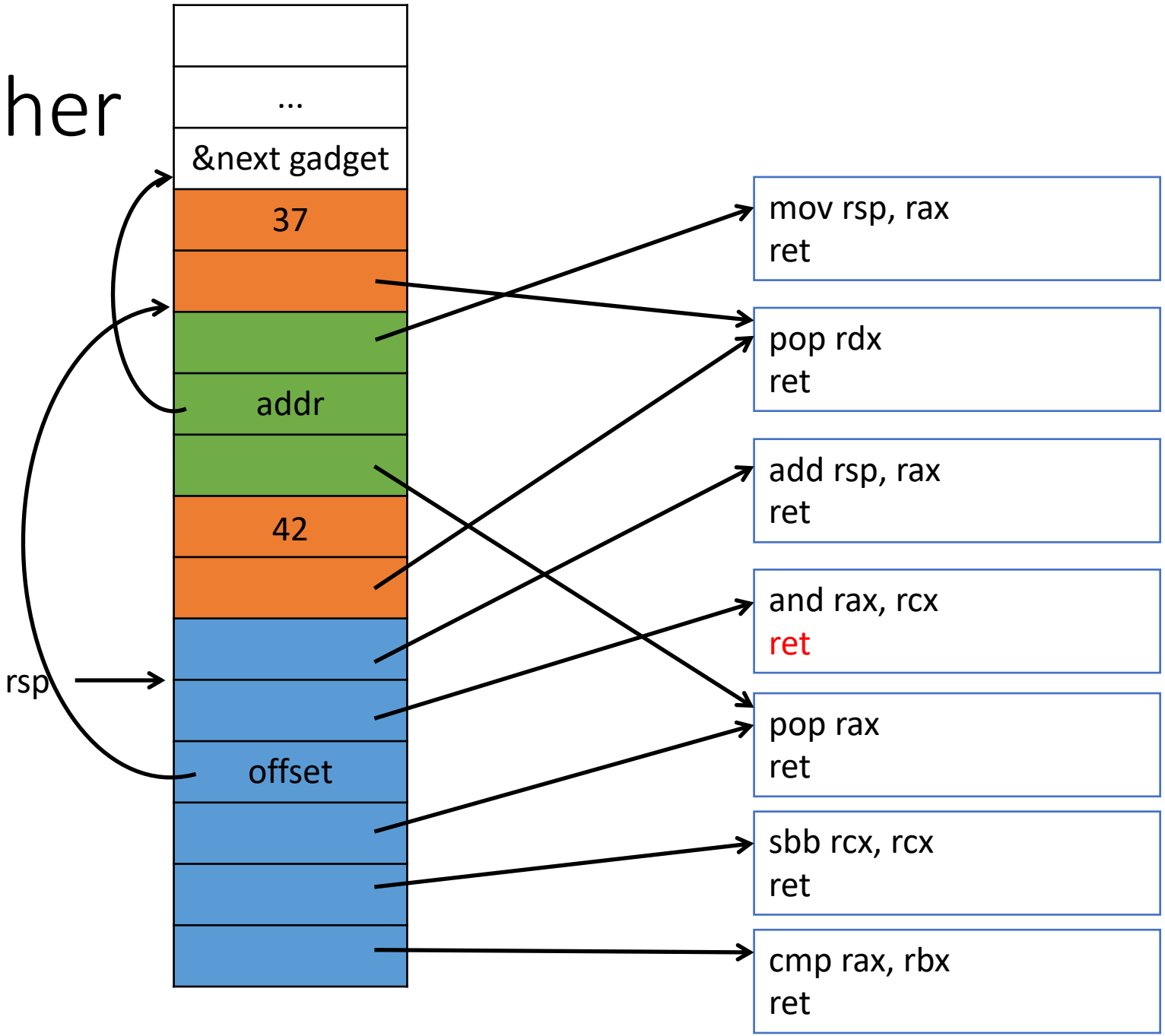
Putting it together

Register	Value
rax	40 = offset
rbx	20
rcx	-1
rdx	17



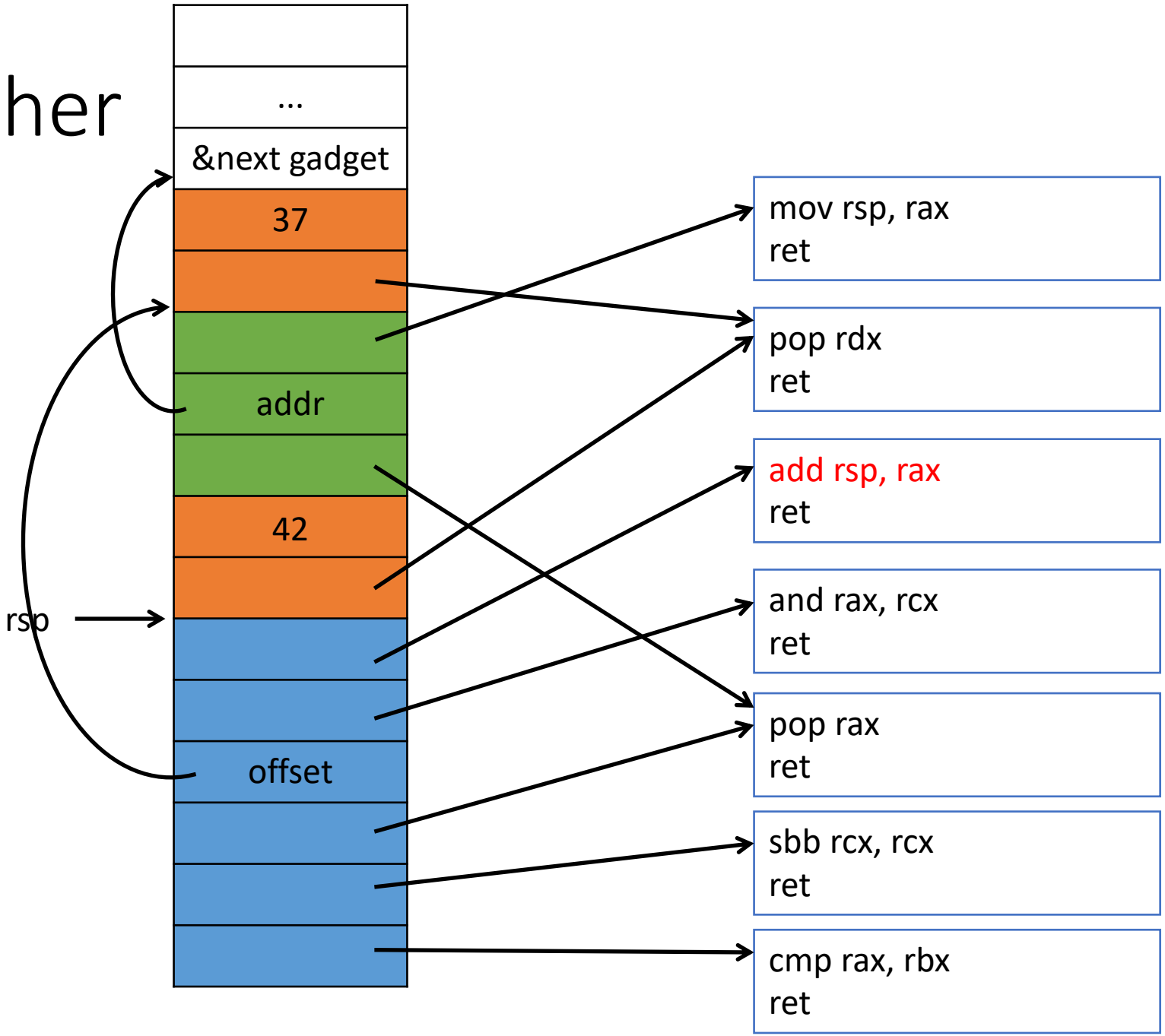
Putting it together

Register	Value
rax	40 = offset
rbx	20
rcx	-1
rdx	17



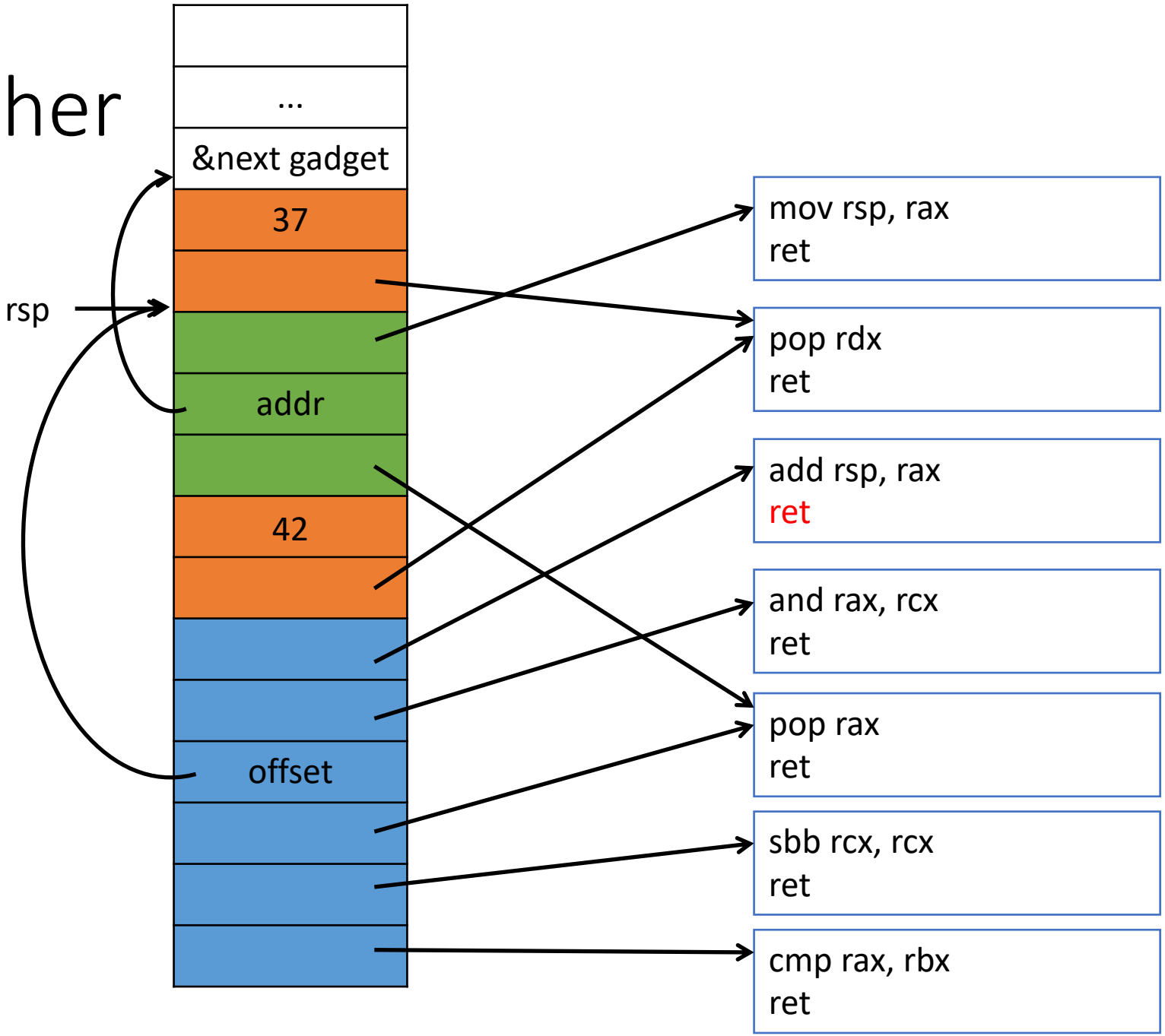
Putting it together

Register	Value
rax	40 = offset
rbx	20
rcx	-1
rdx	17



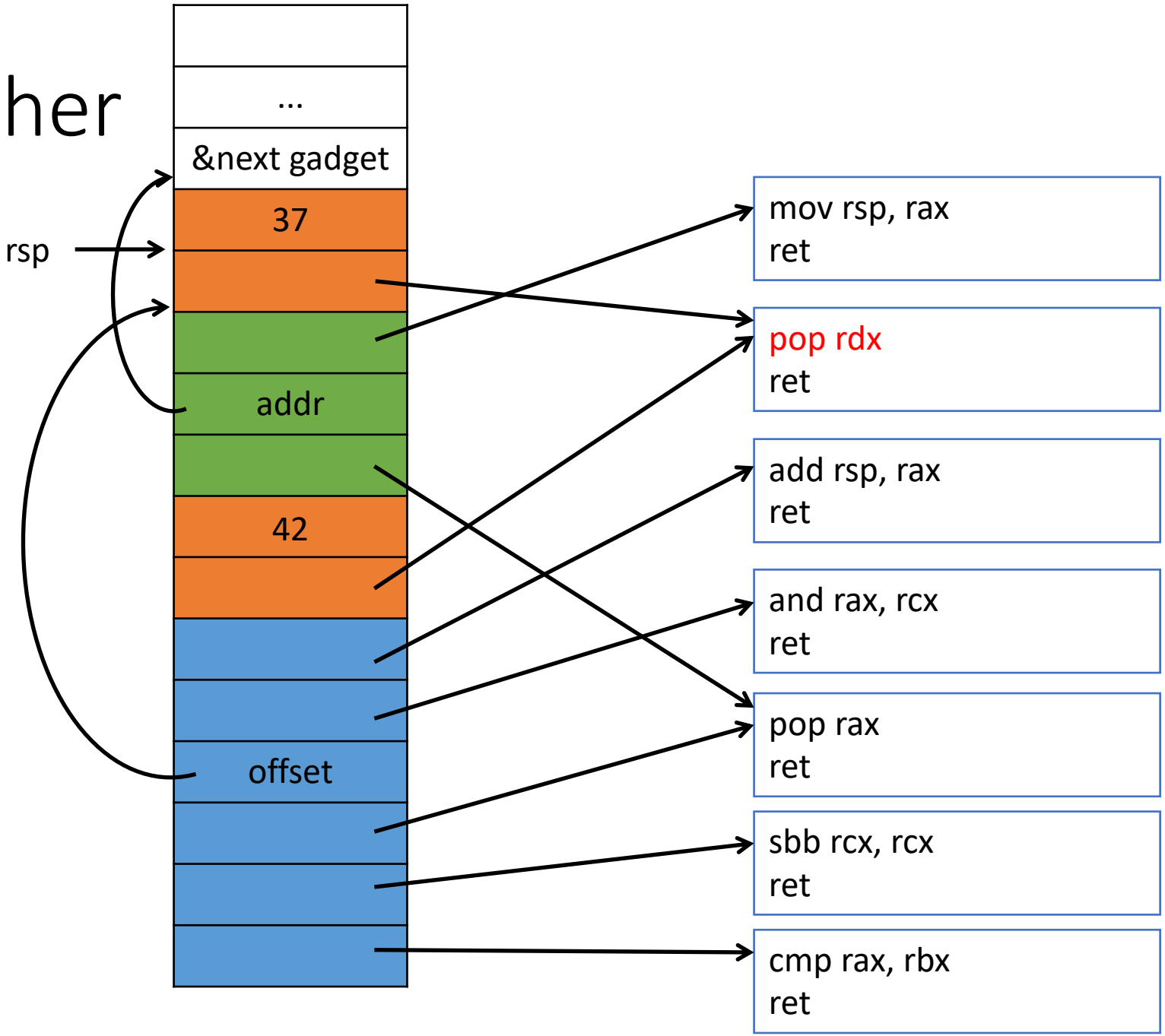
Putting it together

Register	Value
rax	40 = offset
rbx	20
rcx	-1
rdx	17



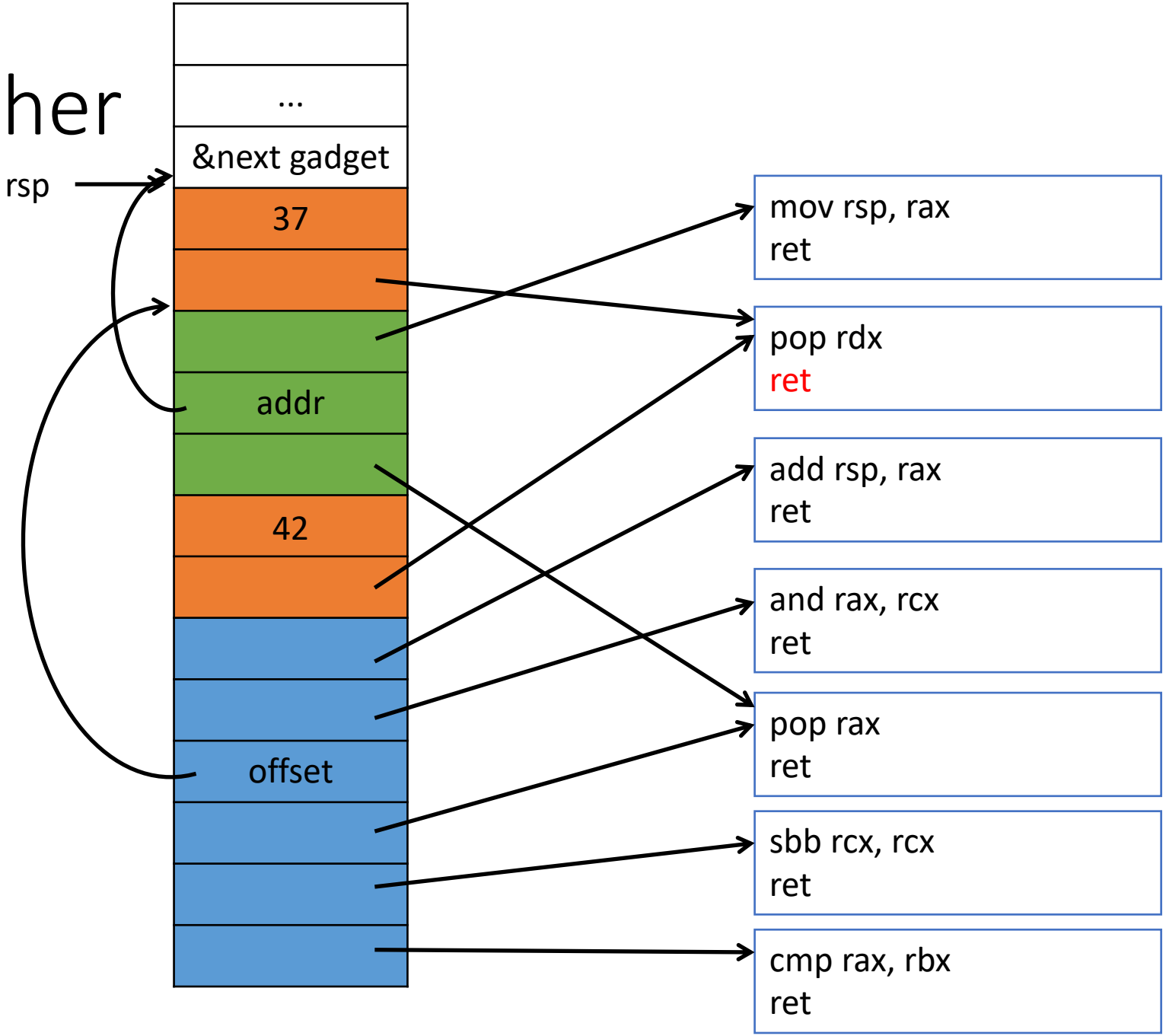
Putting it together

Register	Value
rax	40 = offset
rbx	20
rcx	-1
rdx	17



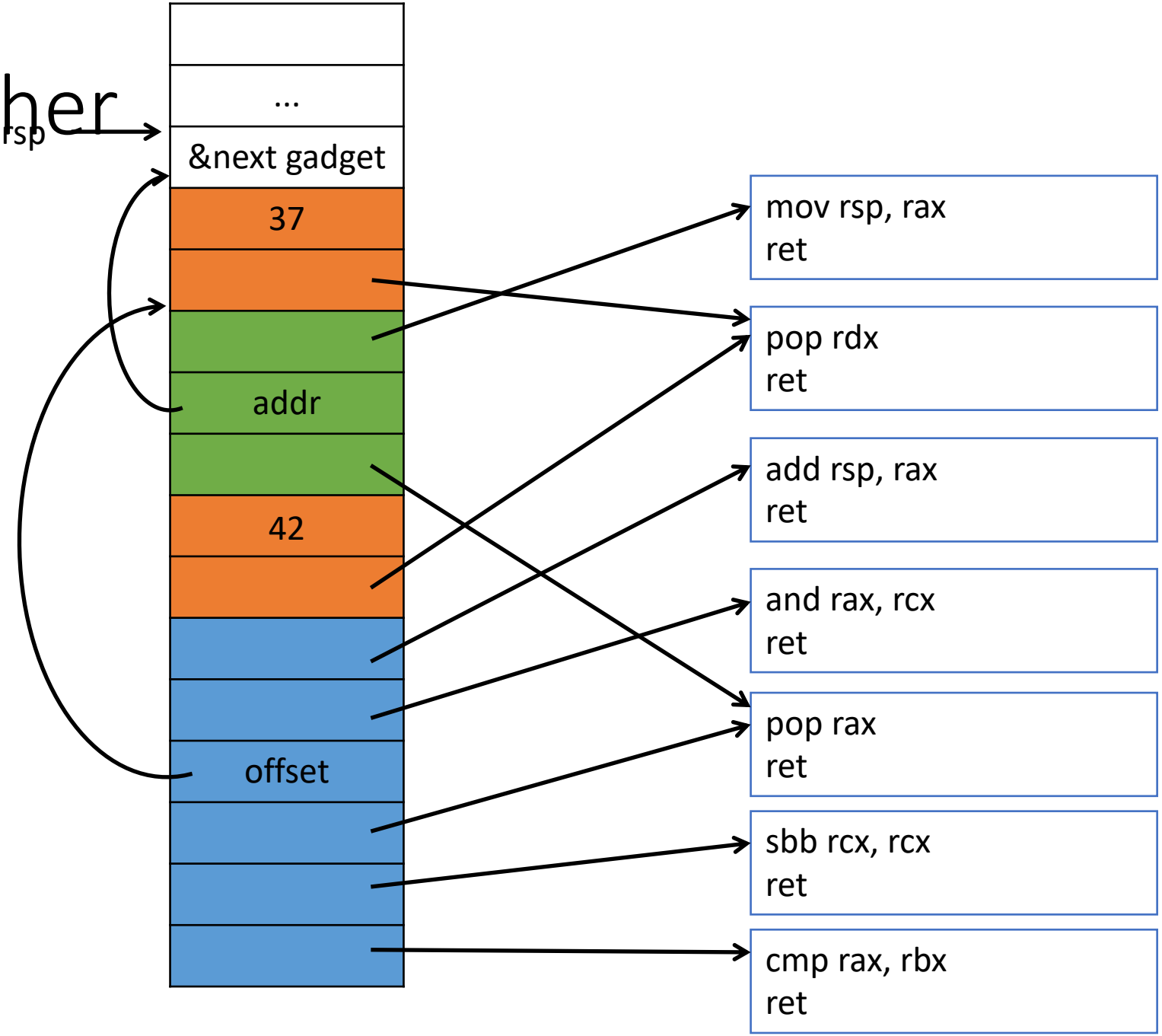
Putting it together

Register	Value
rax	40 = offset
rbx	20
rcx	-1
rdx	37



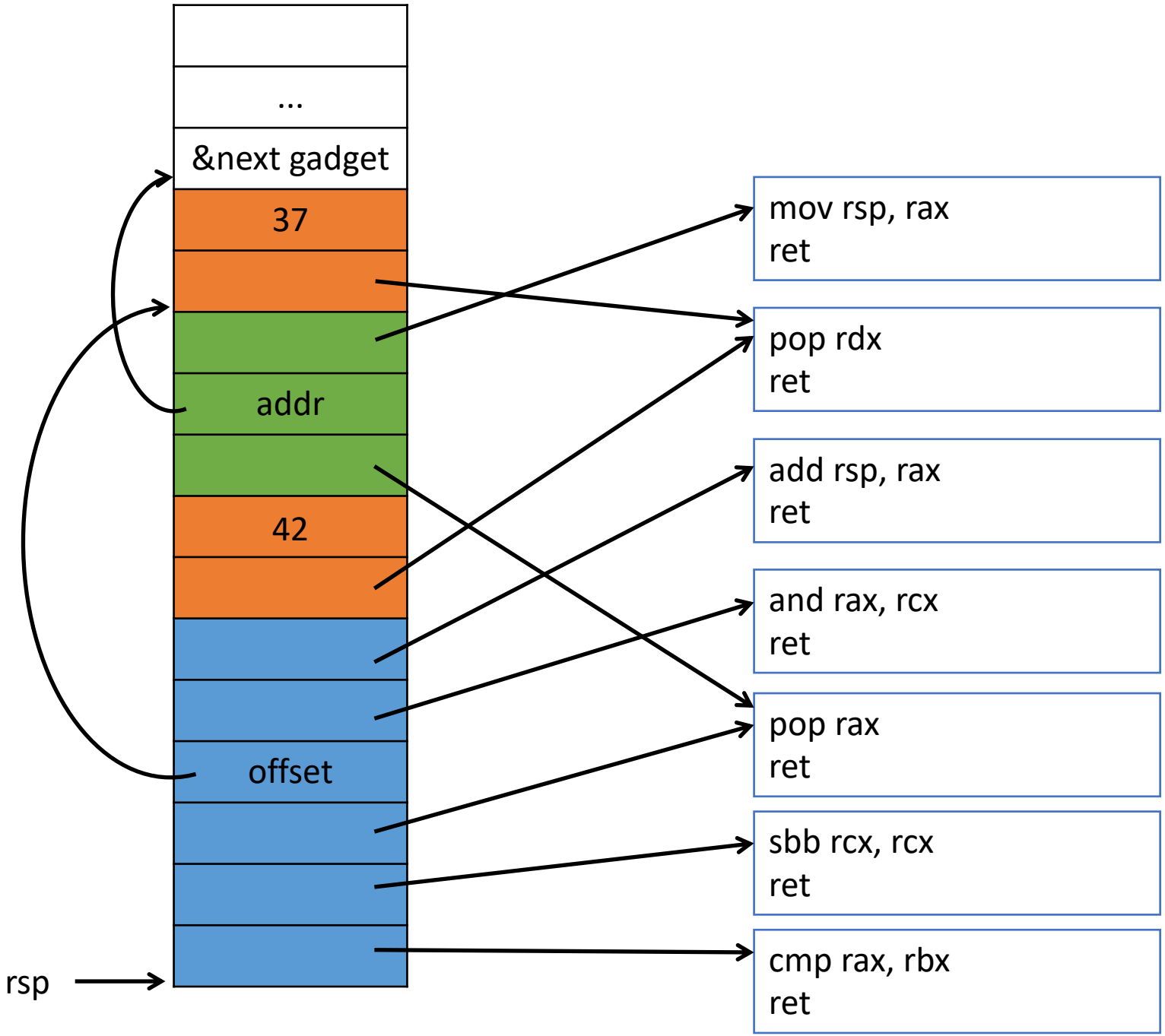
Putting it together

Register	Value
rax	40 = offset
rbx	20
rcx	-1
rdx	37



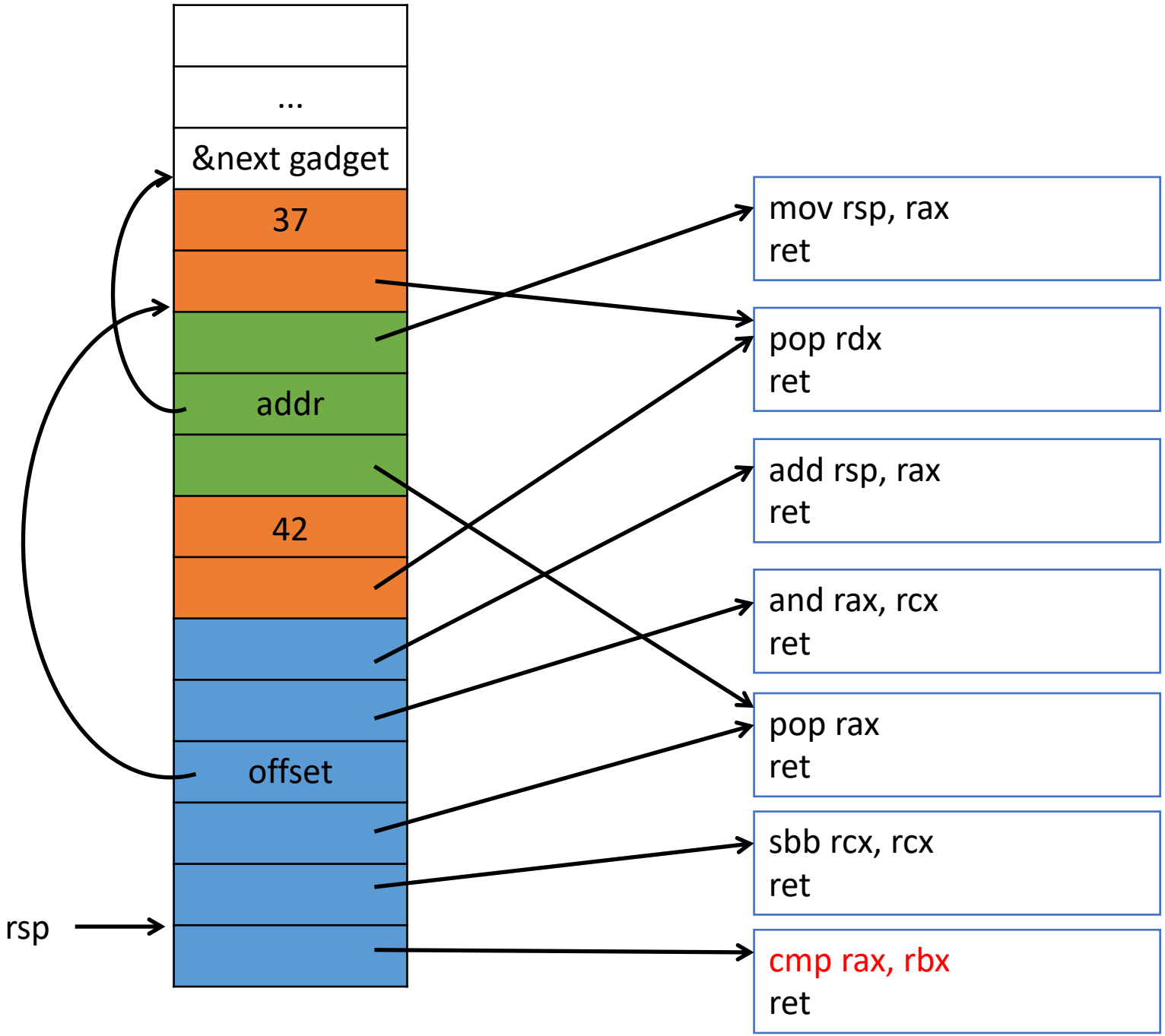
And again!

Register	Value
rax	500
rbx	20
rcx	108
rdx	17



And again!

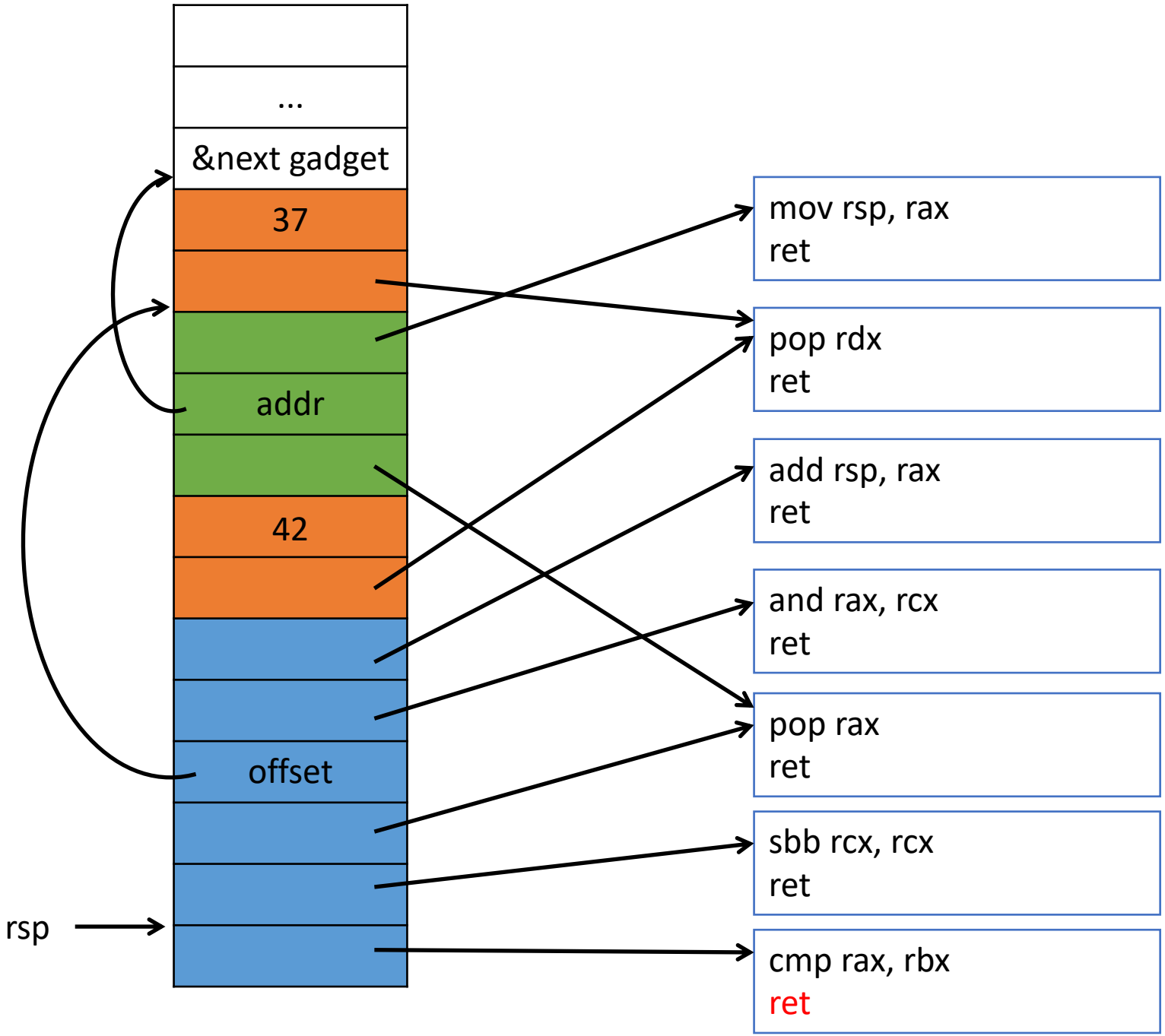
Register	Value
rax	500
rbx	20
rcx	108
rdx	17



And again!

Register	Value
rax	500
rbx	20
rcx	108
rdx	17

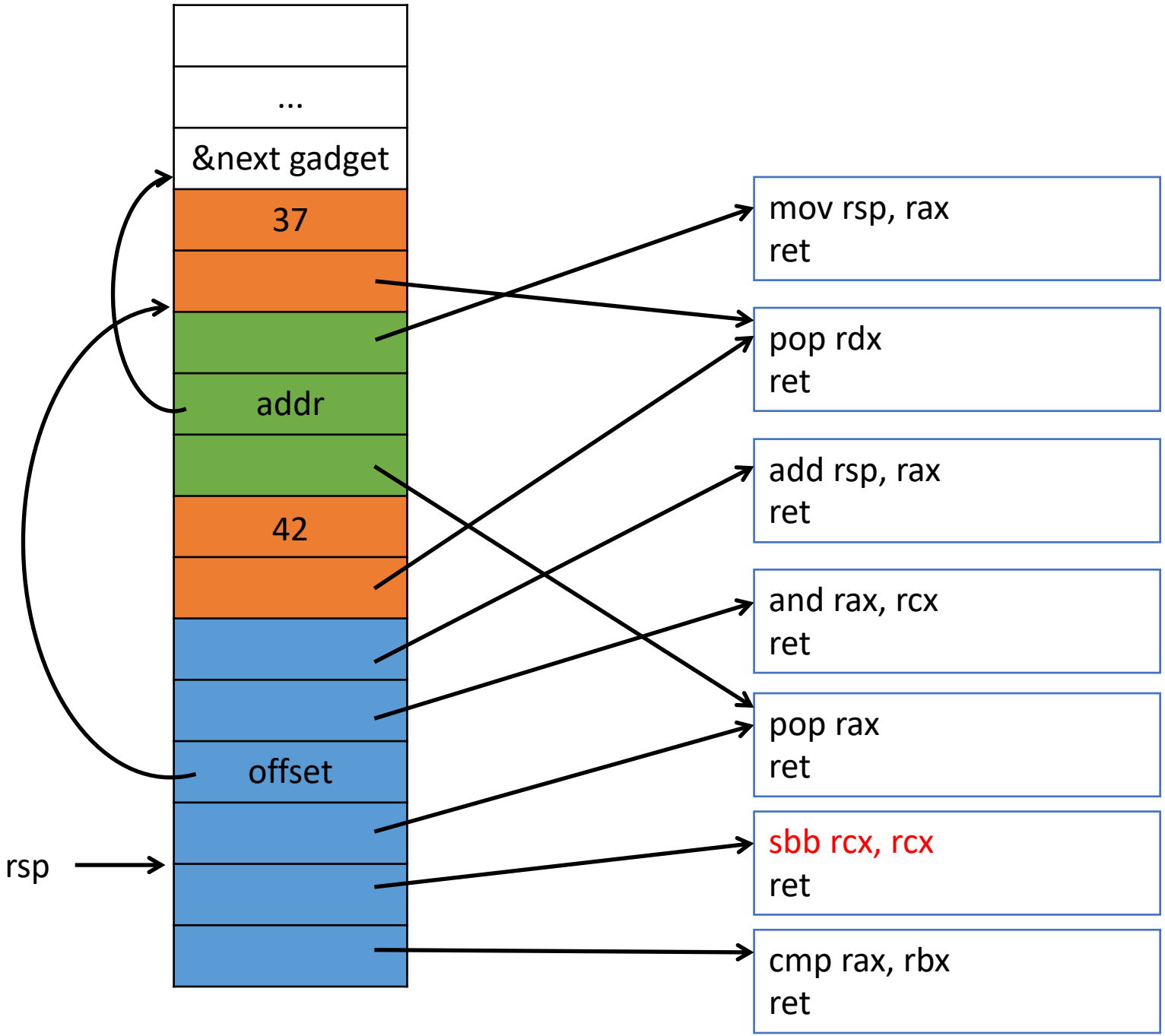
cf = 0



And again!

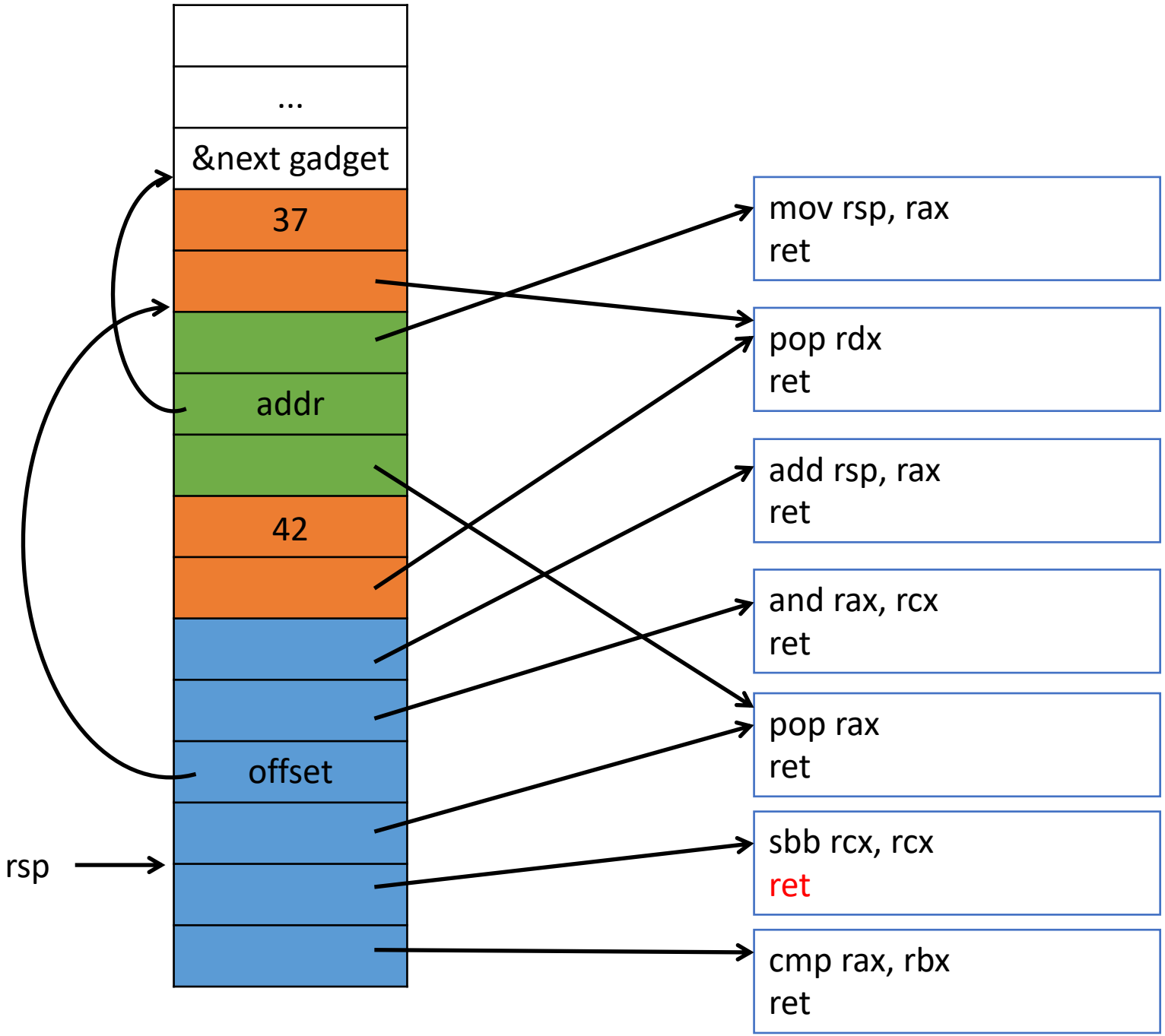
Register	Value
rax	500
rbx	20
rcx	108
rdx	17

cf = 0



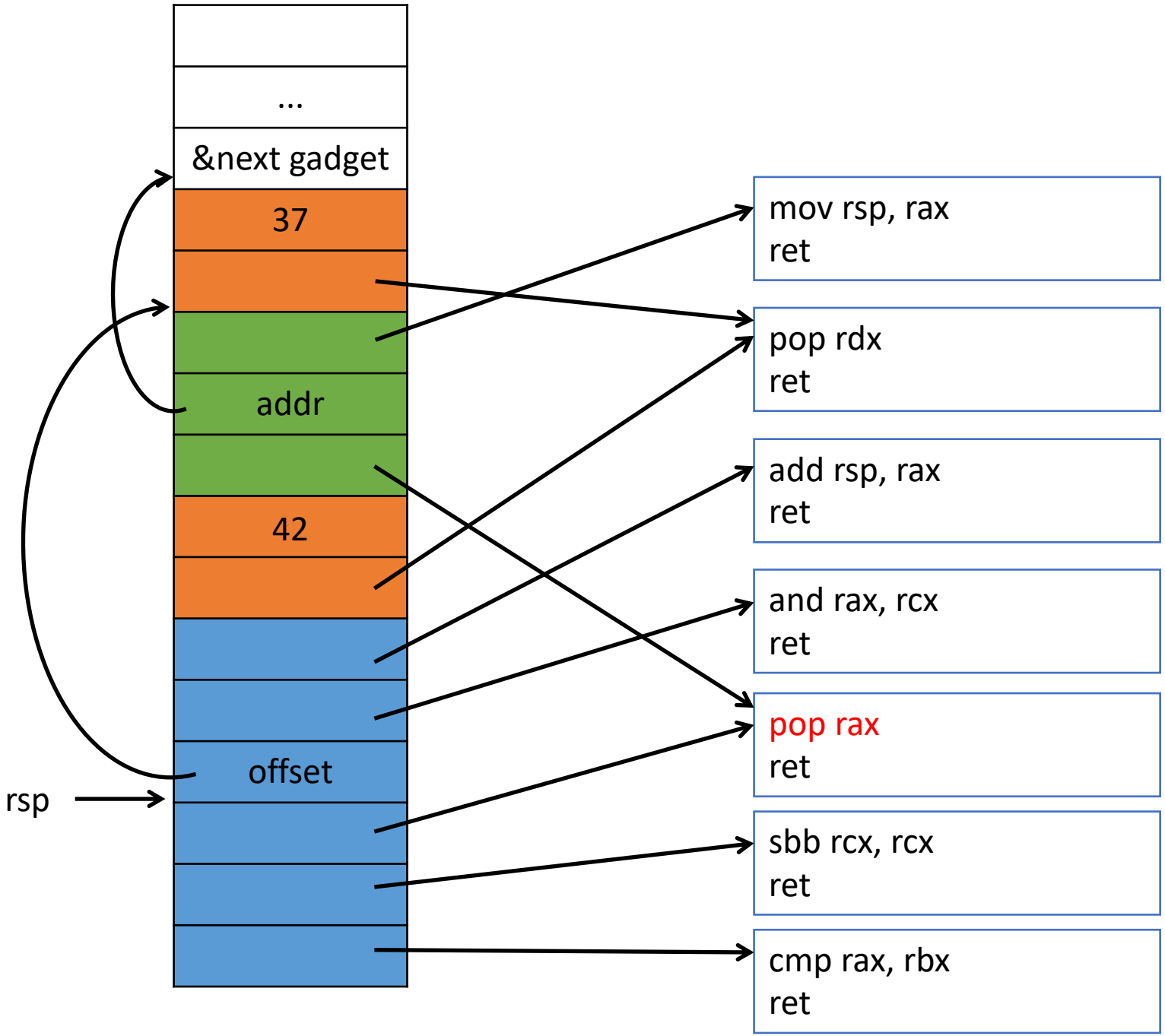
And again!

Register	Value
rax	500
rbx	20
rcx	0
rdx	17



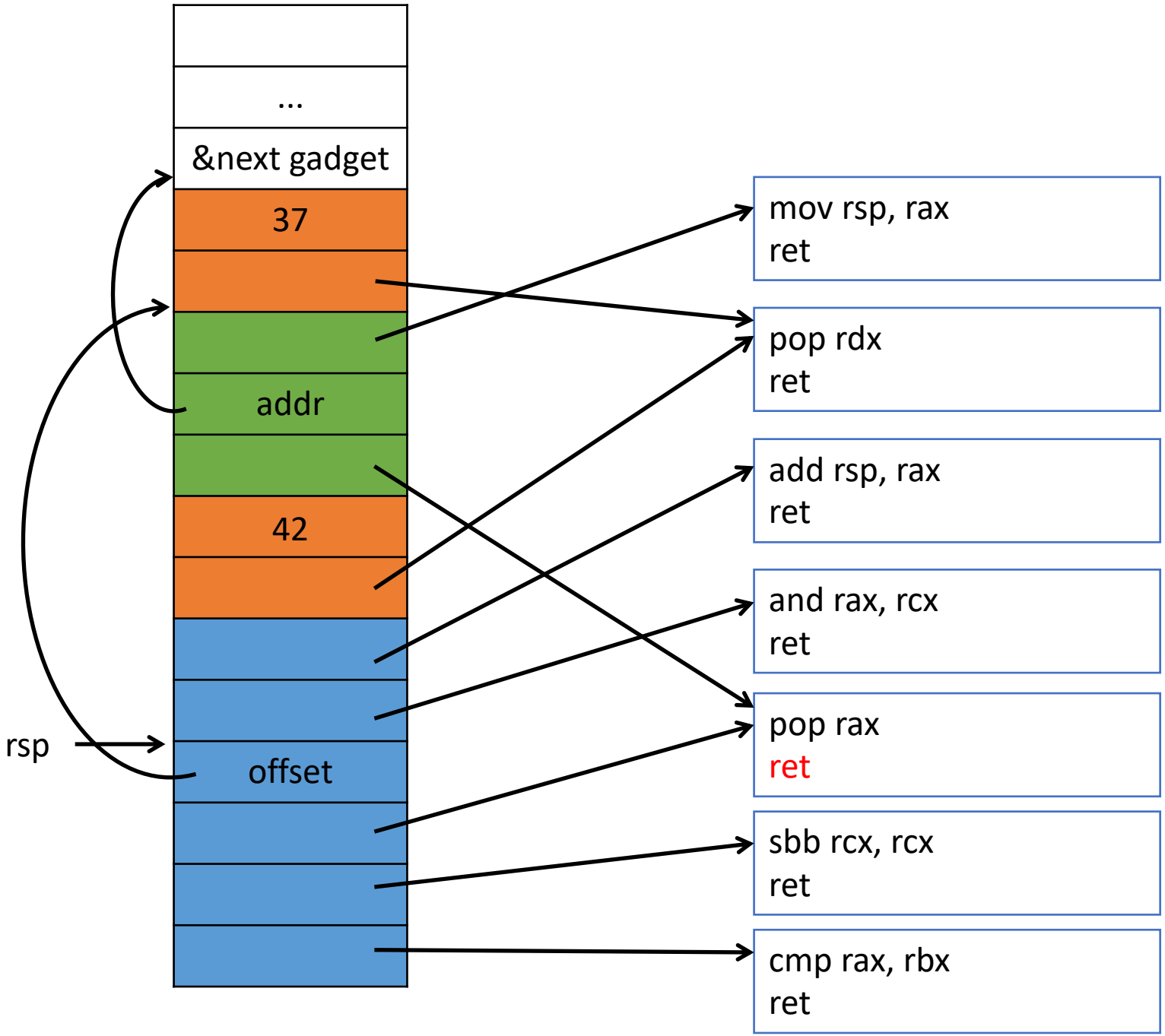
And again!

Register	Value
rax	500
rbx	20
rcx	0
rdx	17



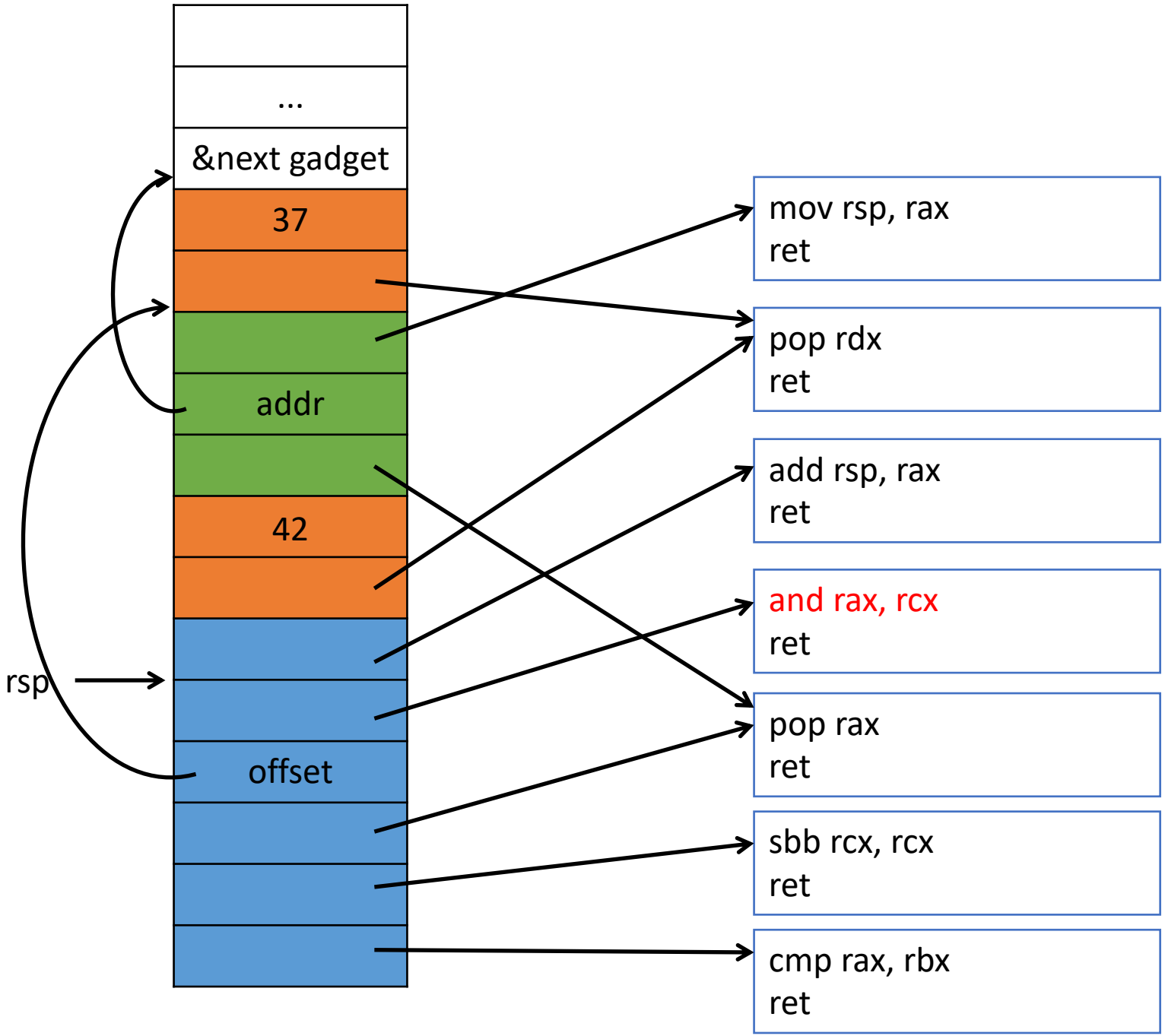
And again!

Register	Value
rax	40 = offset
rbx	20
rcx	0
rdx	17



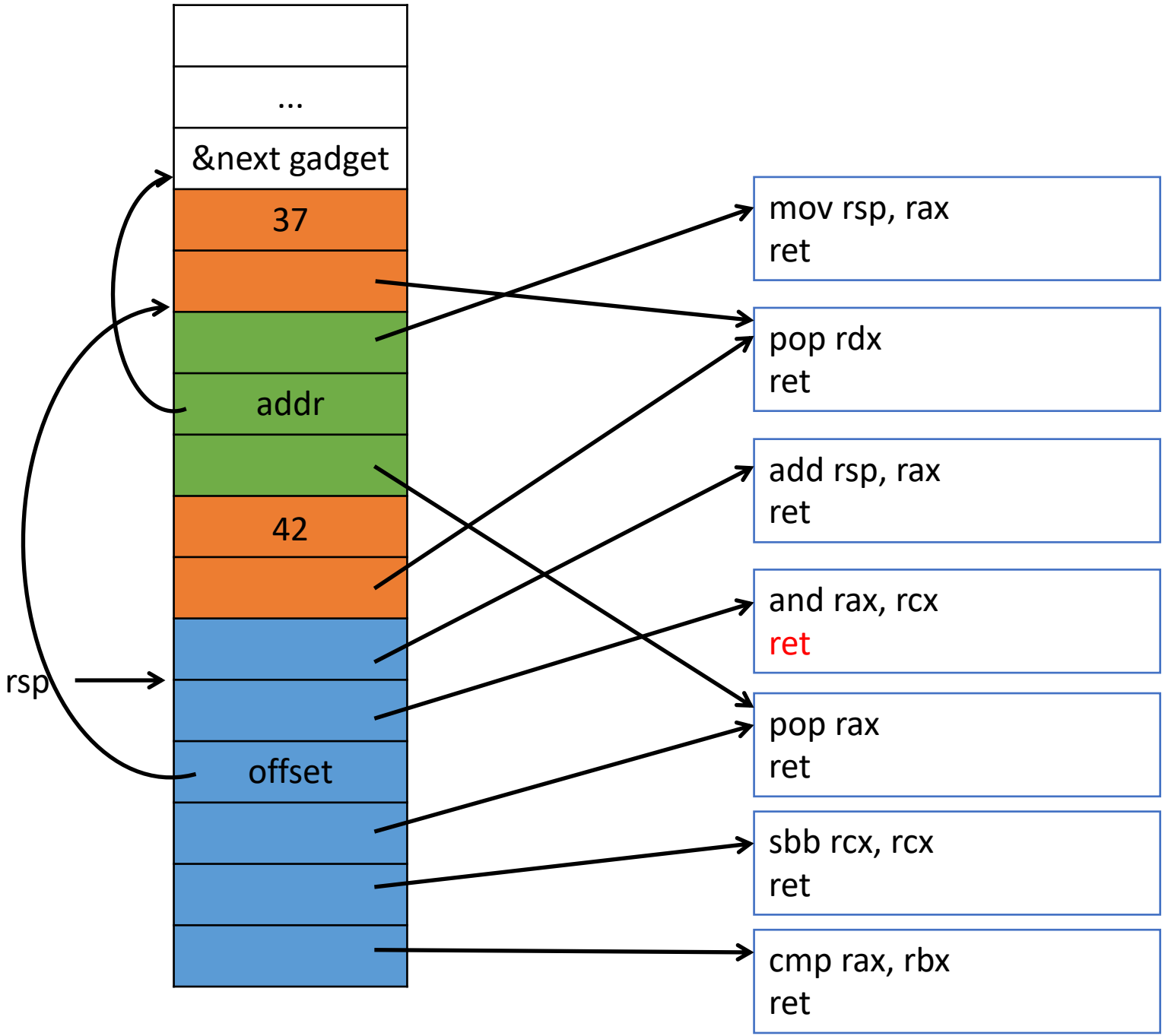
And again!

Register	Value
rax	40 = offset
rbx	20
rcx	0
rdx	17



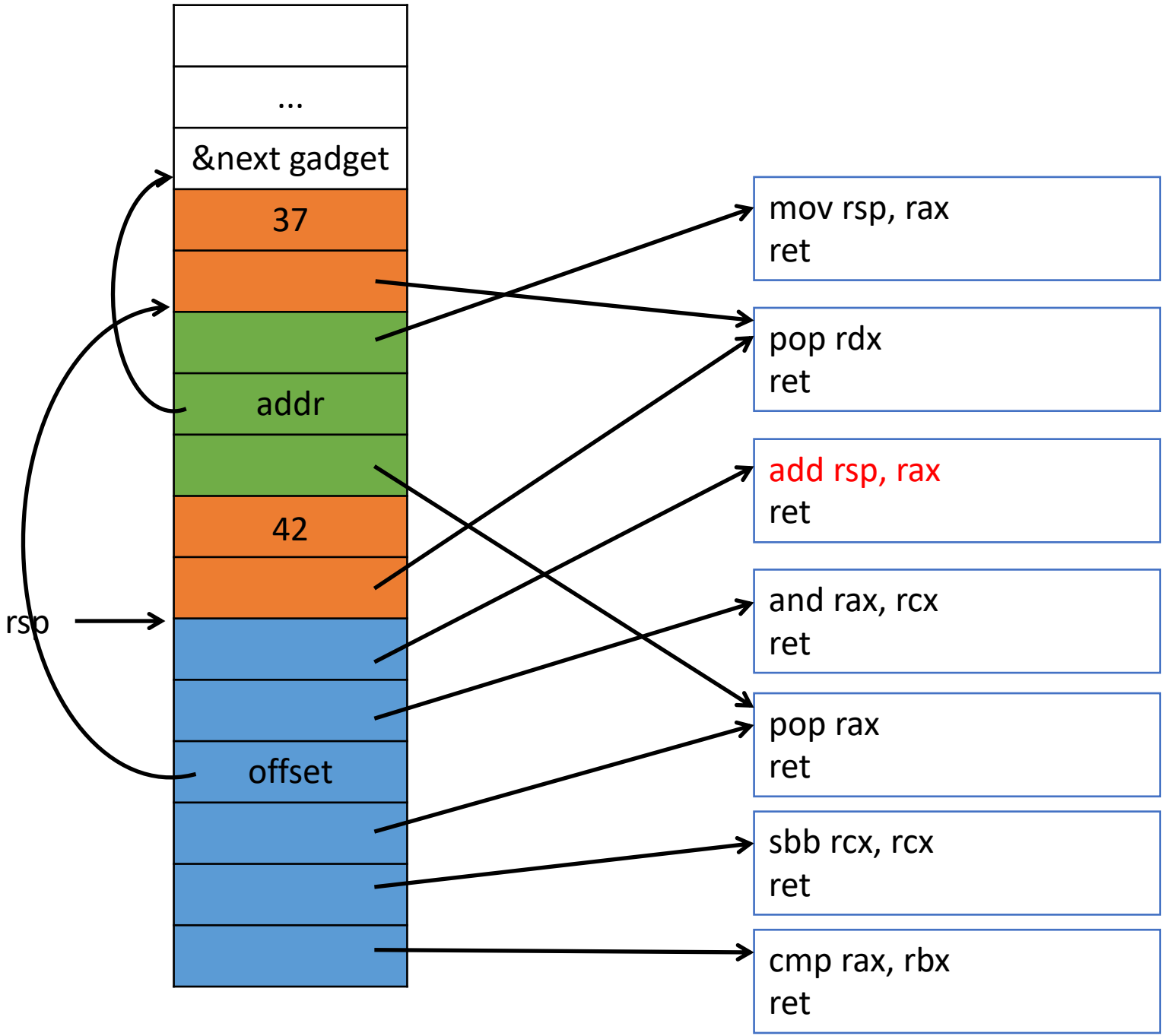
And again!

Register	Value
rax	0
rbx	20
rcx	0
rdx	17



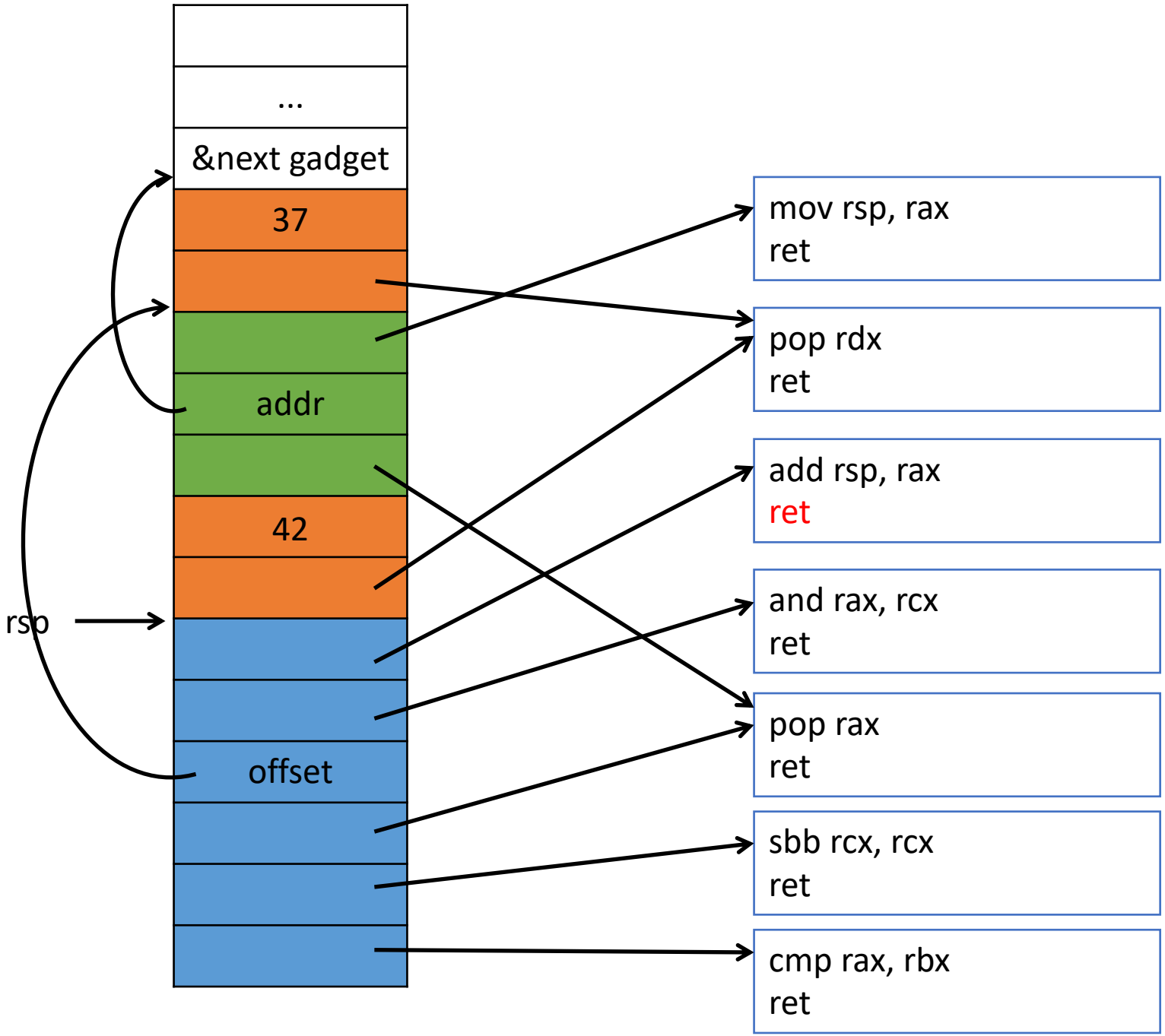
And again!

Register	Value
rax	0
rbx	20
rcx	0
rdx	17



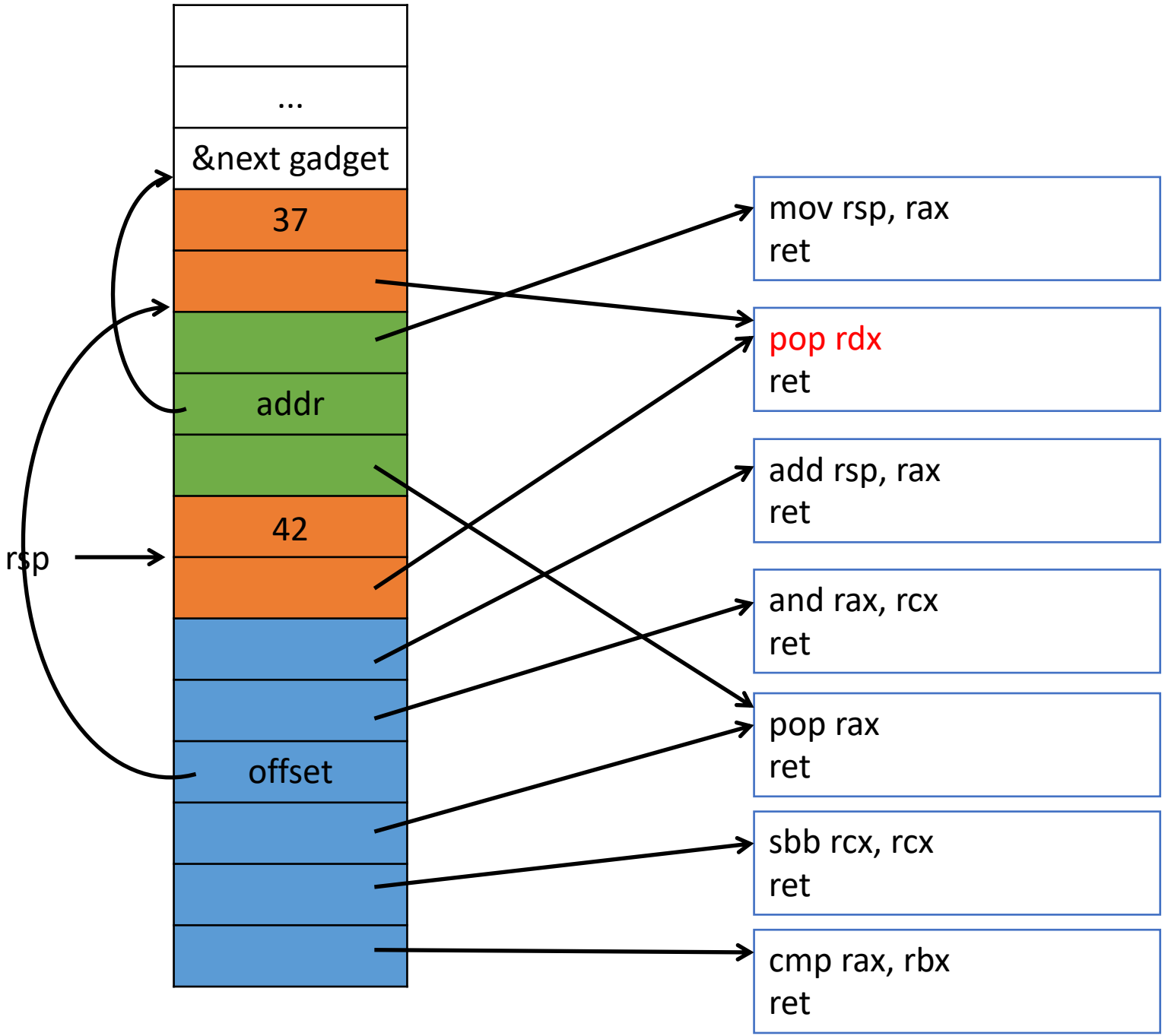
And again!

Register	Value
rax	0
rbx	20
rcx	0
rdx	17



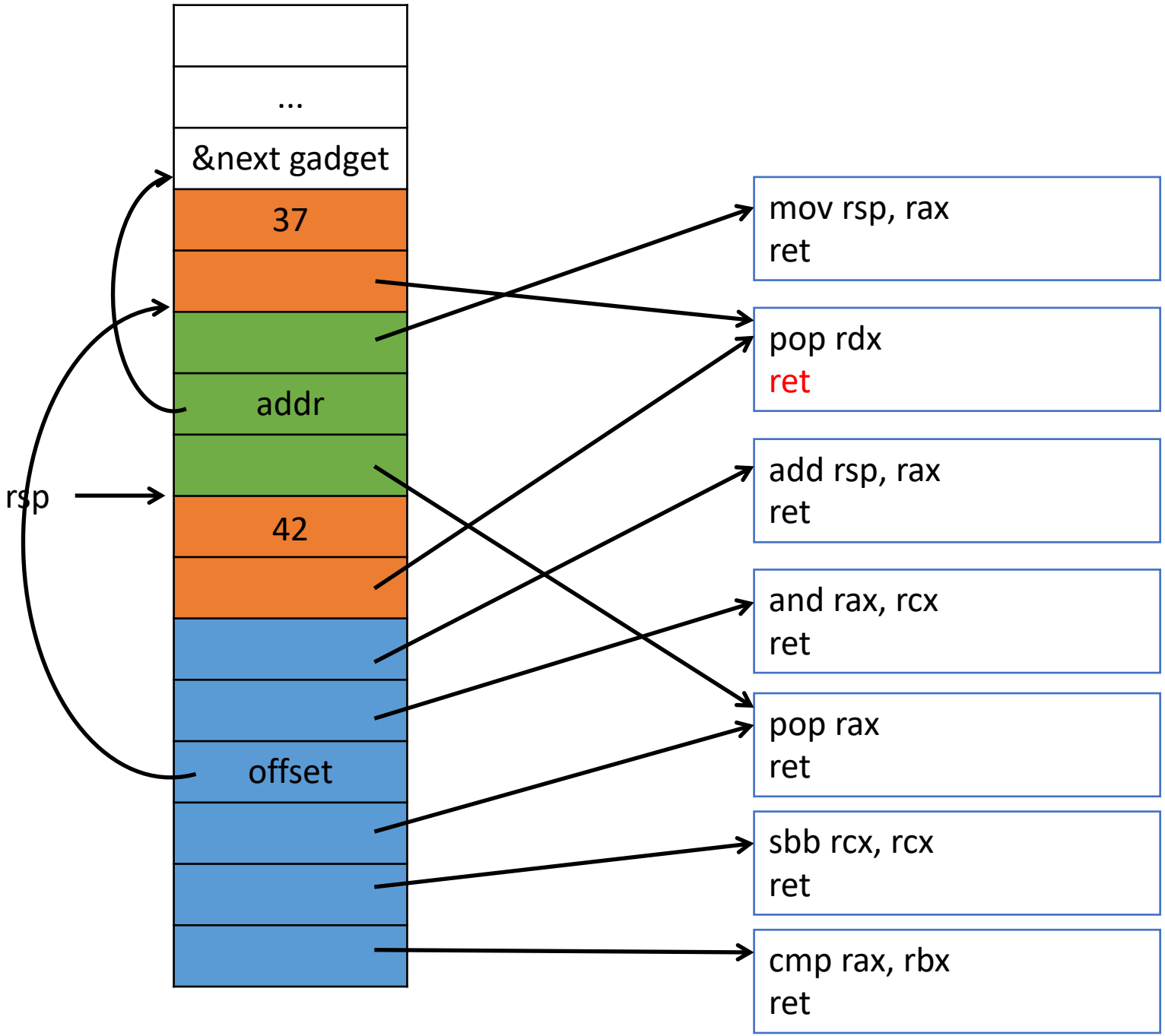
And again!

Register	Value
rax	0
rbx	20
rcx	0
rdx	17



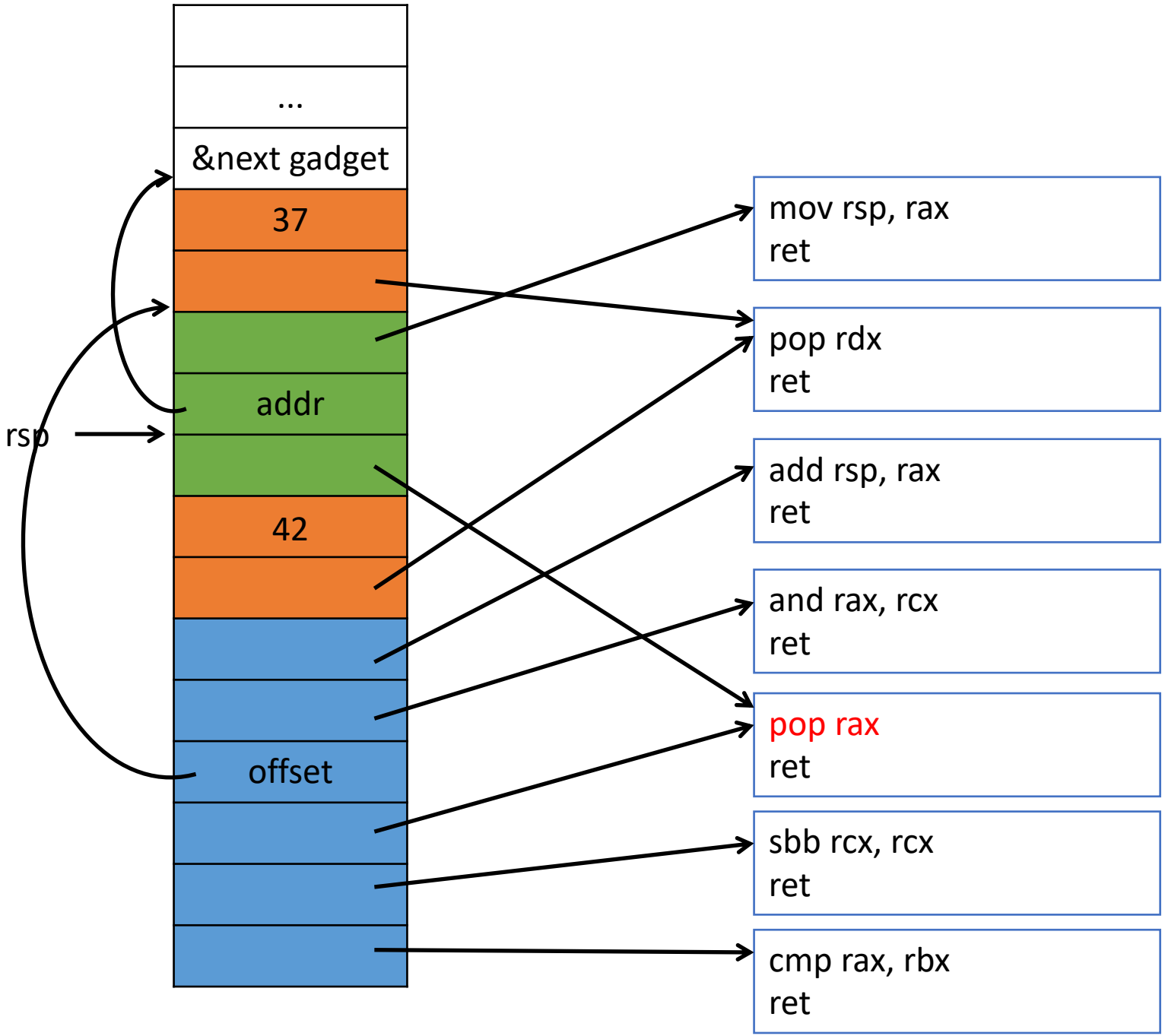
And again!

Register	Value
rax	0
rbx	20
rcx	0
rdx	42



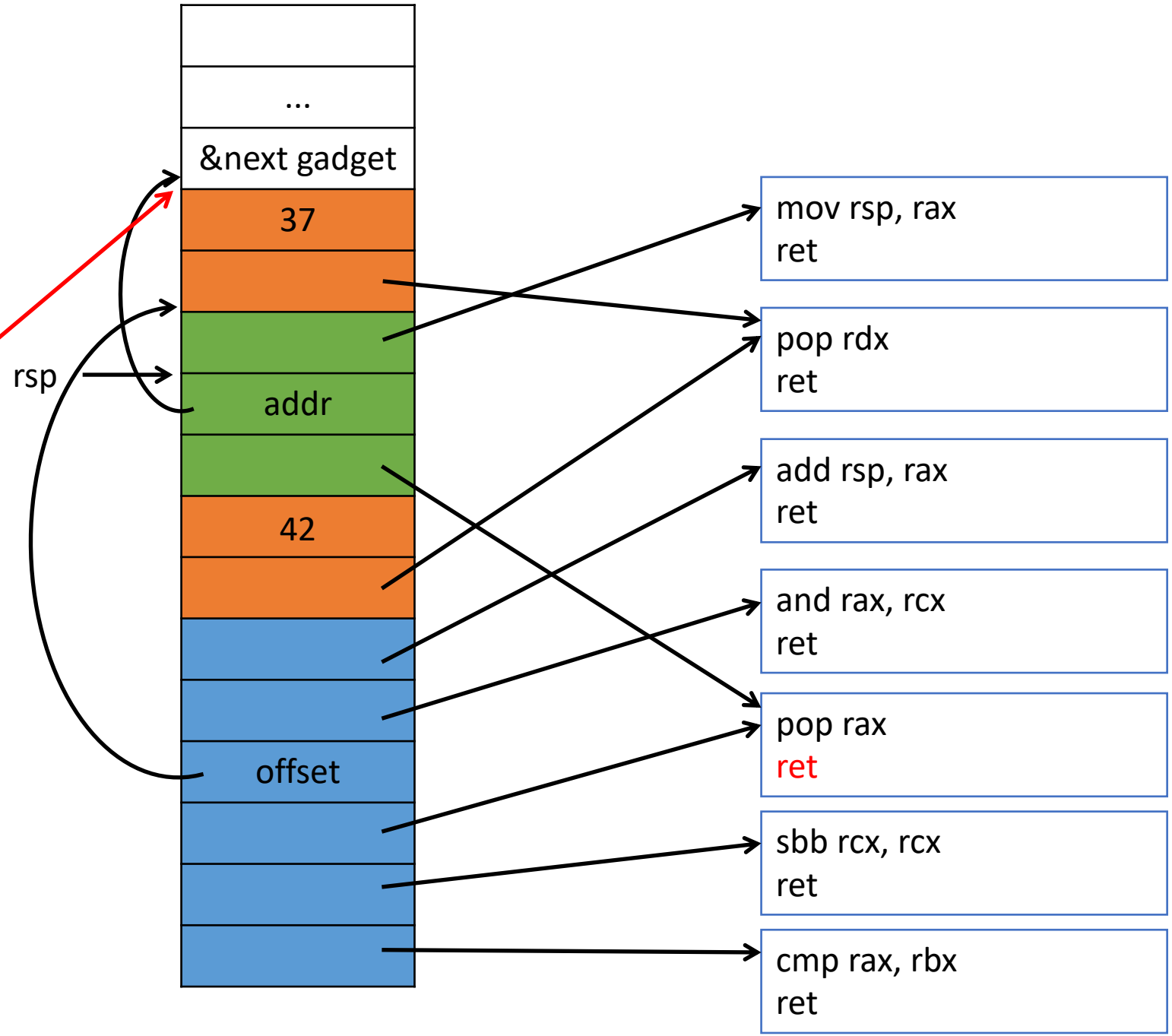
And again!

Register	Value
rax	0
rbx	20
rcx	0
rdx	42



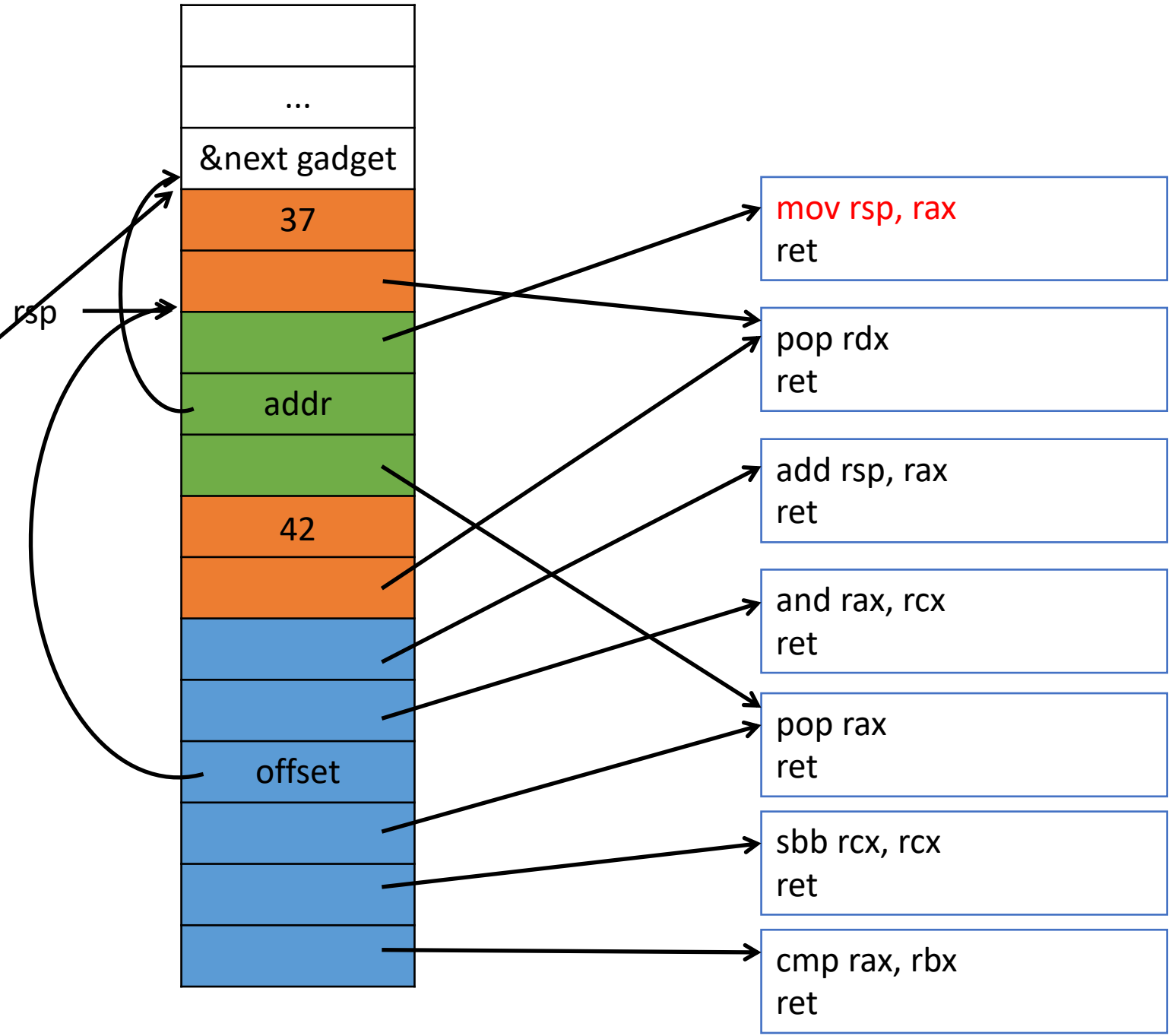
And again!

Register	Value
rax	addr
rbx	20
rcx	0
rdx	42



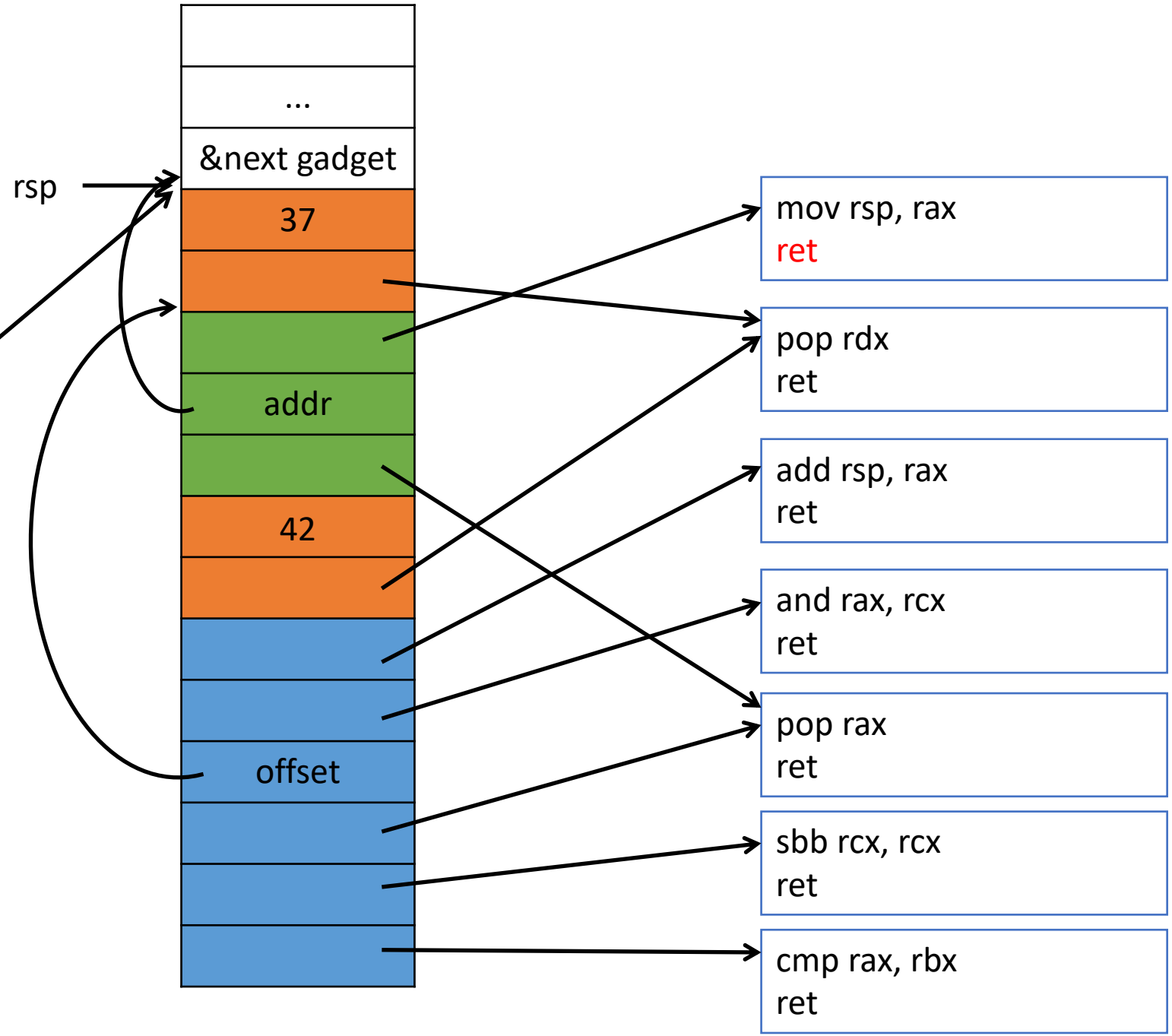
And again!

Register	Value
rax	addr
rbx	20
rcx	0
rdx	42



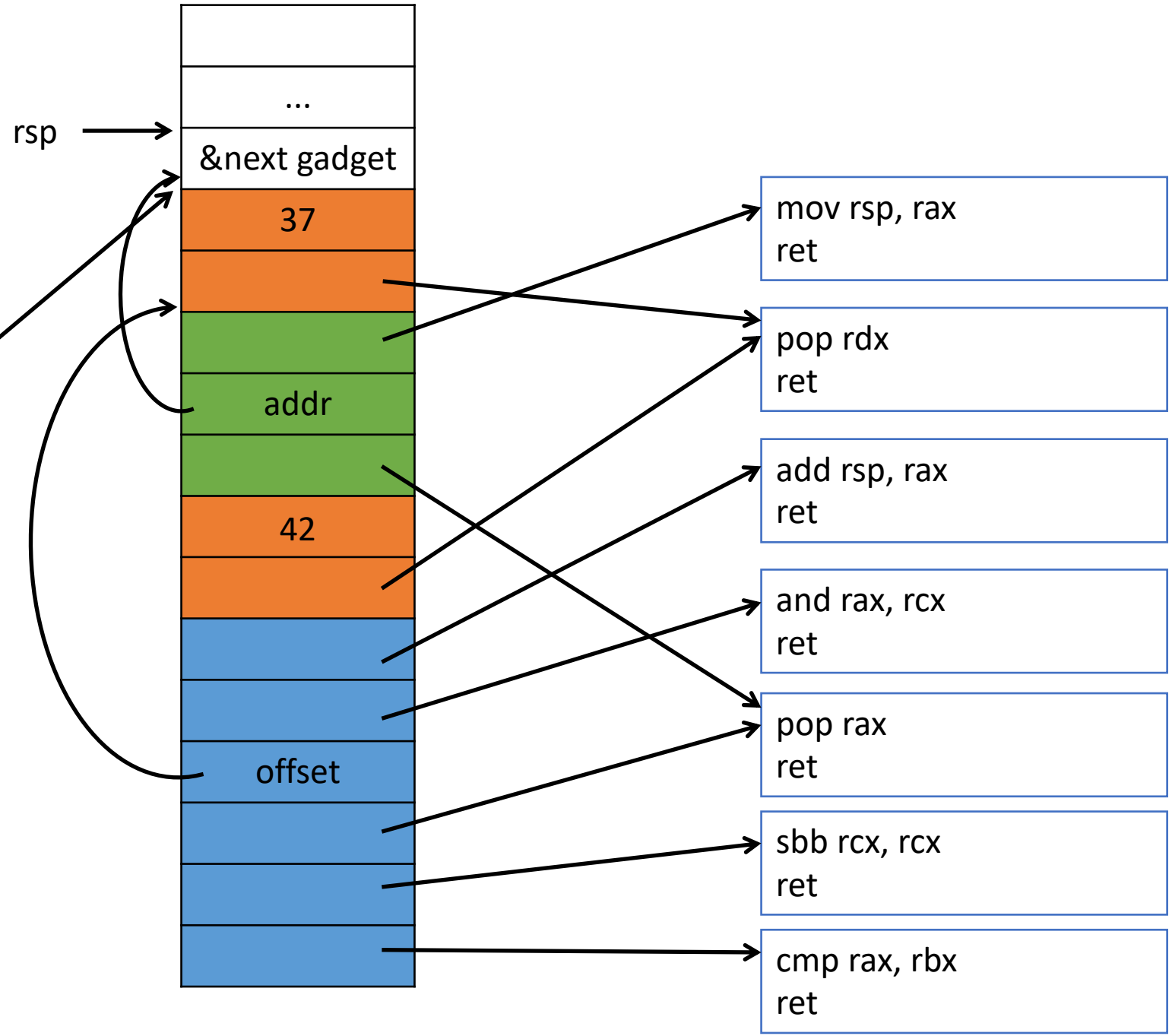
And again!

Register	Value
rax	addr
rbx	20
rcx	0
rdx	42



And again!

Register	Value
rax	addr
rbx	20
rcx	0
rdx	42



Compare

Register	Value
rax	10
rbx	20
rcx	108
rdx	17

Register	Value
rax	40
rbx	20
rcx	-1
rdx	37

```
if (rax < rbx)
        rdx = 37;
else
        rdx = 42;
```

Register	Value
rax	500
rbx	20
rcx	108
rdx	17

Register	Value
rax	addr
rbx	20
rcx	0
rdx	42

Stack pivot: What if the ROP program is not on the stack?

- We can’t always overflow a stack buffer (e.g., canaries)
- Place our return-oriented program somewhere else like a heap-allocated buffer
- Perform a stack pivot: change rsp to that buffer
  - Requires overwriting the saved return address or a code pointer with the address of a pivot gadget

Some pivoting options

- xchg rax, rsp ; ret exchanges the values in rax and rsp
- push rax ; pop rsp ; ret
- Also useful if you don’t know stack addresses but do know the address of where the return-oriented program is written

More code = more ROP options

- The more code in programs and libraries, the easier it is to construct return-oriented programs
- If you cannot find the gadget you need directly, you can often use multiple gadgets to construct the same behavior
- Sometimes you'll need to deal with extra instructions between the one you care about and the return
- Sometimes those instructions have requirements
  - E.g., `mov [rax + 16], rdx ; div [rcx] ; ret` requires you put the address you want minus 16 in rax, rcx needs to be a valid address pointing to a nonzero value, and rax and rdx will both change as a result!

Other gadget endings

- We’ve looked at sequences of instructions ending in ret; not the only option
- `pop rdx ; ret 16` will pop a value into rdx, pop the next word into rip, and then increment the stack pointer by 16
- `mov rax, rbx ; jmp rcx` works just fine as a gadget so long as rcx holds the address of a ret instruction
- You can even construct a whole return-oriented program that never uses a ret, e.g., by finding gadgets ending in pop rax ; jmp rax
- Far return: retf pops rip and cs off the stack

Other architectures

- ROP works on basically every computer architecture
  - x86, x86-64, ARM32, ARM64, RISC-V, MIPS, Sparc, Z80
- Some architectures require less code or are easier than others (in my experience)
  - x86 and ARM32 are easy
  - Z80 required a tiny amount of code (every instruction sequence is valid)
  - x86-64 requires more code
  - Sparc has a weird register window calling convention
  - RISC-V is “annoying” I’ve been told